

Results — revised submission — original style

1 Datasets

#	Name	N	P	K	MeanIR	CVIR
1	GpositivePseAAC	519	440	4	3.86	1.15
2	Emotions	593	72	6	1.48	0.16
3	Scene	2407	294	6	1.25	0.11
4	PlantPseAAC	978	440	12	6.69	0.68
5	HumanPseAAC	3106	440	14	15.29	1.05
6	Yeast	2417	103	14	7.20	1.82
7	birds	645	260	19	5.41	0.80
8	medical	978	1449	45	222.37	0.72
9	enron	1702	1001	53	10.45	0.80

Table 1: Datasets used in the revised experiments. N , P , K are the number of instances, features, labels. MeanIR and CVIR report per-label imbalance (higher = more imbalanced).

Pending follow-ups (to fill before final submission)

- **ECC baseline on birds (RF + LGBM).** The `birds` folders contain only PA/PR/BR/CC/CLR — ECC has not been run. Method slot 12 currently shows as a gap on every `birds` panel. Re-run with `scripts/train_extra_baselines.py --dataset birds --algorithm ecc` for both `--base_learner rf` and `--base_learner lgbm`.
- **LGBM run on enron.** Sequentially training (2–3h per task due to $K=53$ pairwise classifiers); will be merged into `full_enron_split_lgbm_summary/` when complete and the LGBM aggregate panels regenerated.
- **Extended PartialAbstention metrics.** The evaluator has been extended with 11 new PA metrics (`jaccard_pa`, `{example,macro,micro}-{precision,recall,f1}_pa`, `mfrd_pa`, `afrd_pa`). Existing runs were summarised with the previous metric set; the new columns will populate automatically on the next end-to-end training+evaluation pass.

2 Method encoding

For every panel that follows, the x-axis encodes the classifier as one of 12 methods, grouped into three blocks:

Block A — Order-based predictors (ours, indices 1–8). Each name has the form `<order>-<target>-<height>`; $order \in \{\text{PA} = \text{partial order}, \text{PR} = \text{pre-order}\}$; $target \in \{\text{H} = \text{Hamming}, \text{S} = \text{Subset 0/1}\}$; $height \in \{2, \infty \text{ (omitted in the label)}\}$.

- 1 = **PA-H-2**: partial order, Hamming target, height ≤ 2 .
- 2 = **PA-H**: partial order, Hamming target, unrestricted height.
- 3 = **PA-S-2**: partial order, Subset 0/1 target, height ≤ 2 .
- 4 = **PA-S**: partial order, Subset 0/1, unrestricted.
- 5 = **PR-H-2**: pre-order, Hamming, height ≤ 2 .
- 6 = **PR-H**: pre-order, Hamming, unrestricted.
- 7 = **PR-S-2**: pre-order, Subset 0/1, height ≤ 2 .
- 8 = **PR-S**: pre-order, Subset 0/1, unrestricted.

Block B — Standard MLC baselines from the original paper (indices 9–11).

- 9 = **BR** (Binary Relevance): K independent binary classifiers, one per label.
- 10 = **CC** (Classifier Chain): sequential chain that uses previous labels as features.
- 11 = **CLR** (Calibrated Label Ranking): pairwise label ranking with a calibration label that separates relevant from irrelevant.

Block C — Extended baseline added for the revision (index 12).

- 12 = **ECC** (Ensemble of Classifier Chains): averages over multiple CCs with random chain orders.

Reading the panels. Each panel plots one dotted line with markers per noise level, colored as in the original paper’s Table 5: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$, $\alpha = 0.3$. *PartialAbstention* and *ScoreVector* panels show only indices 1–8 (the remaining baselines do not export a partial-abstention prediction nor a probabilistic score vector). Vertical dashed lines at $x = 4.5$, $x = 8.5$, and $x = 11.5$ separate the four method groups: PA (1–4), PR (5–8), standard baselines BR/CC/CLR (9–11), and ECC (12). Missing methods show as gaps in the line (with a small grey “x” on the axis when the method has no data at all for that panel).

3 Main results: RF base learner, aggregated

RF • Aggregate • BinaryVector — Standard MLC predictions

RF • Aggregate • PartialAbstention — Partial abstention

RF • Aggregate • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

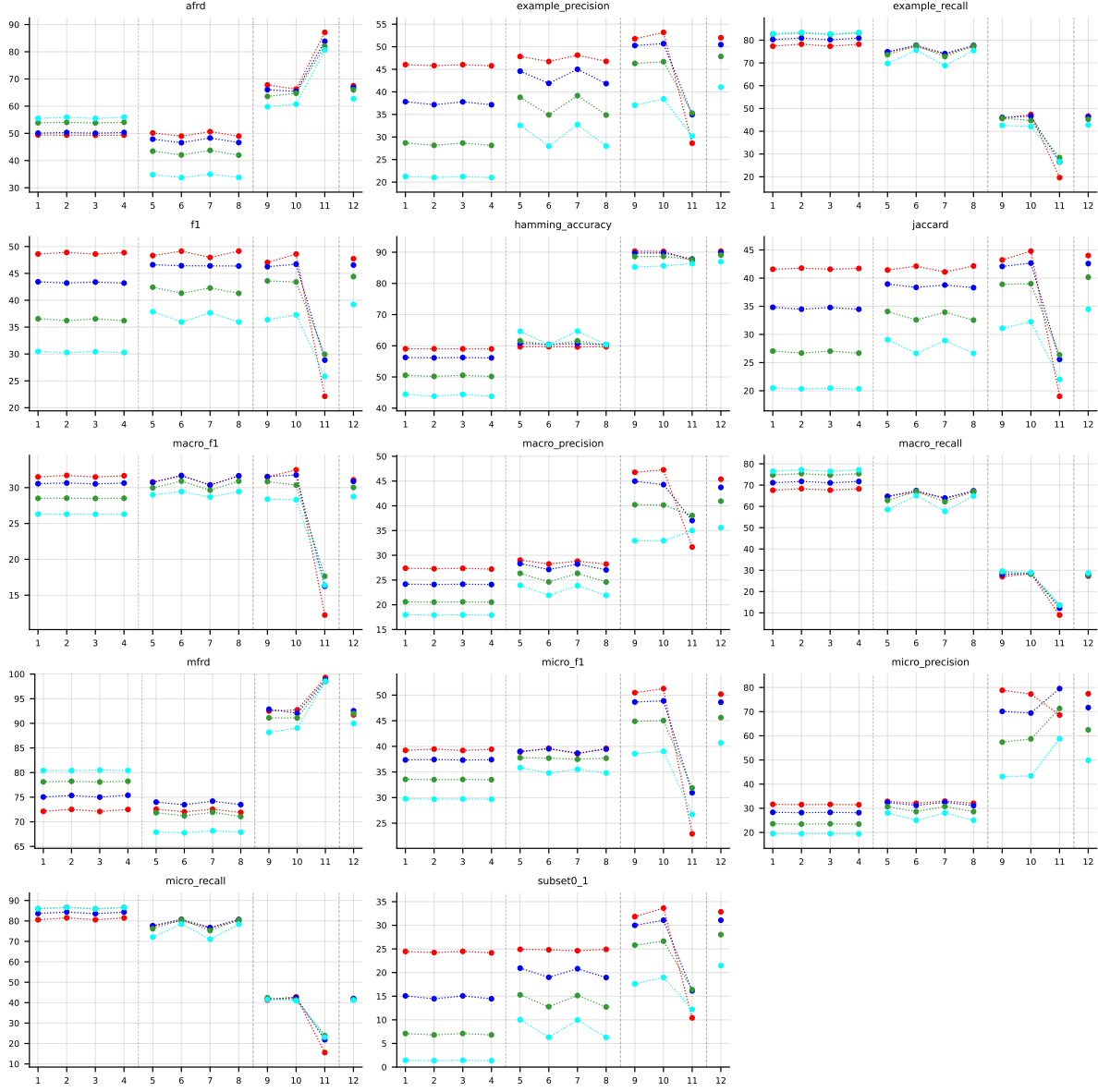


Table 2: Standard MLC predictions (BinaryVector), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

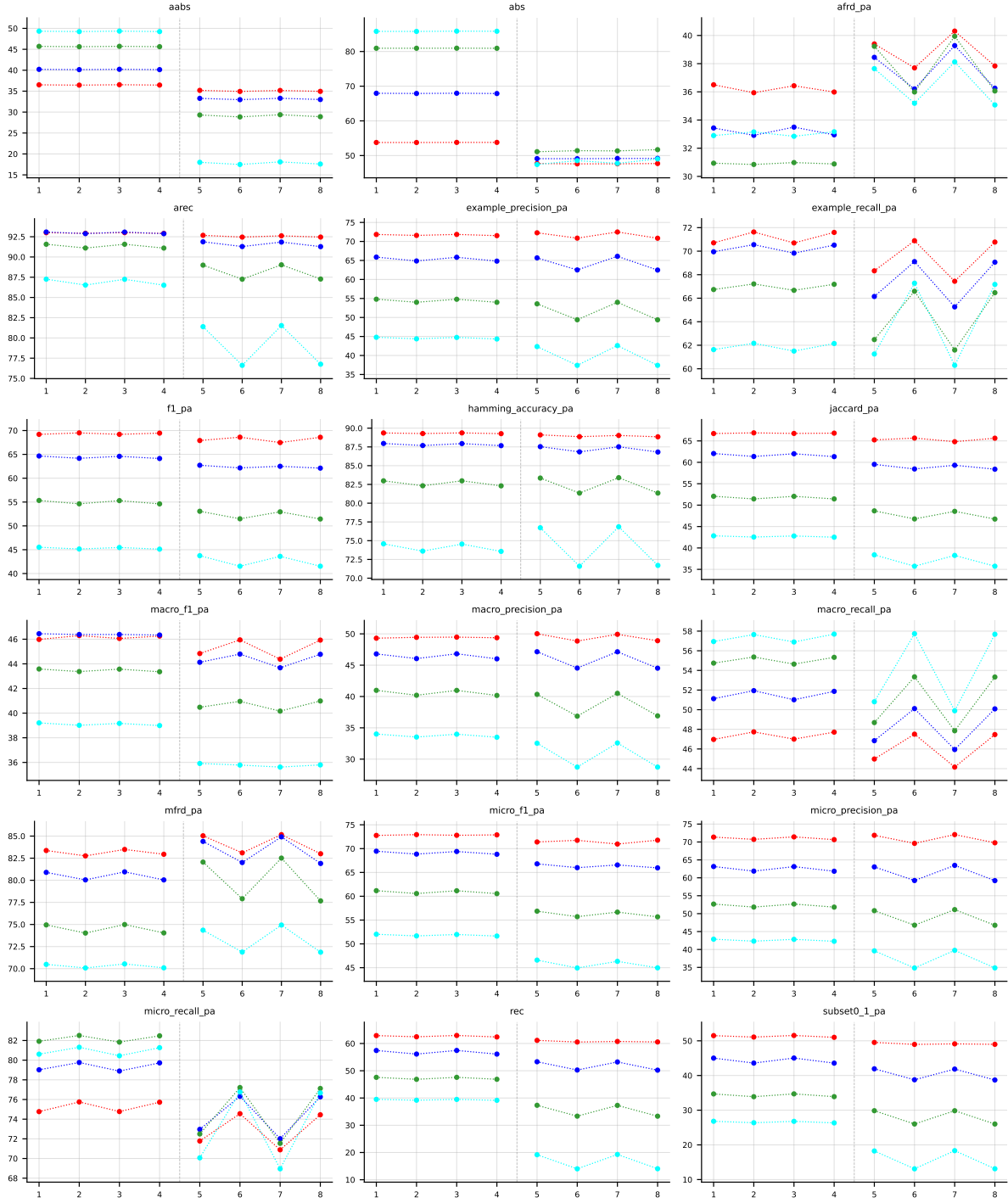


Table 3: Partial abstention (PartialAbstention), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

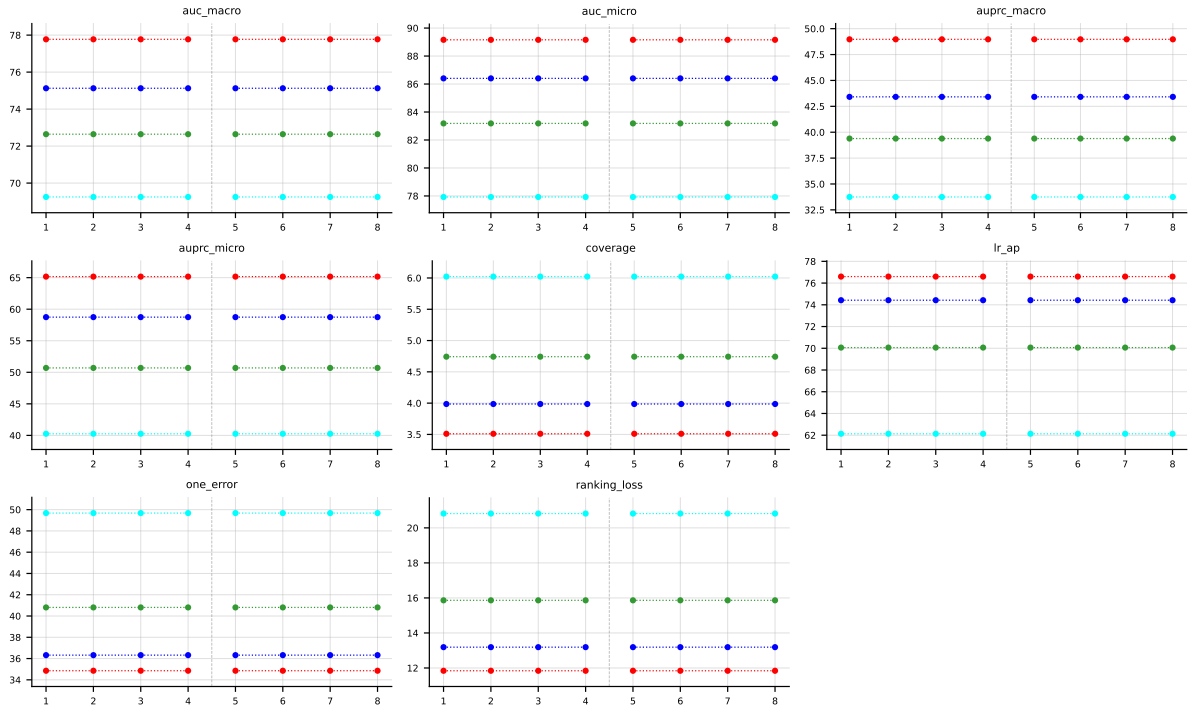


Table 4: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

4 Abstention vs standard MLC

Why this section exists. The paper’s central claim is that an order-based predictor can *abstain* on labels it is unsure about and, on the labels it does predict, achieve higher quality than a standard MLC method that is forced to predict every label. The two chart families below make that claim visible.

Terminology. ‘*Standard MLC*’ (a.k.a. **BinaryVector** / BV) means each method outputs a 0/1 prediction for every one of the K labels. ‘*With abstention*’ (a.k.a. **PartialAbstention** / PA) means an order-based predictor may return $\perp$ (abstain) on some labels; the PA quality metrics (**f1_pa**, **jaccard_pa**, ...) are computed only over the non-abstained labels. Only the 8 PA/PR methods (indices 1–8) can abstain; BR/CC/CLR/ECC cannot.

Reading the coverage-risk chart. Each point is one (PA/PR method, noise level α) pair. The x-axis is the abstention rate (fraction of labels the method skipped); the y-axis is the quality metric on the labels that *were* predicted (higher is better). Marker shape encodes method, colour encodes noise level. The dashed grey line is the best standard-MLC baseline (BR / CC / CLR / ECC) on the BV quality metric — it is the bar that “no abstention” has to clear. **Any point above the dashed line is direct evidence that abstaining improved quality beyond what any standard MLC method achieves.** Points further right traded coverage for that gain.

Reading the paired-bars chart. Same data, different view. The four sub-panels correspond to the four noise levels. Inside each sub-panel: 12 method positions, two bars per position — blue is the standard-MLC quality (**f1**, no abstention), warm-colour is the with-abstention quality on retained labels. Only the 8 PA/PR methods (1–8) have a warm bar; the standard MLC baselines (9–12) only have a blue bar by construction. The green “+ $x.y$ ” label above each warm bar is the gain in score-points over that method’s own BV bar.

4.1 RF, metric f1_pa — aggregate



Figure 1: RF (f1_pa) averaged across all datasets: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).



Figure 2: RF ($f1_{pa}$) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).



Figure 3: RF ($f1_{pa}$) averaged across all datasets — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

4.2 RF, metric $f1_{pa}$ — per dataset

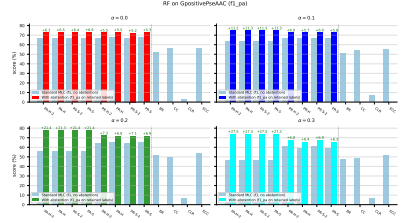
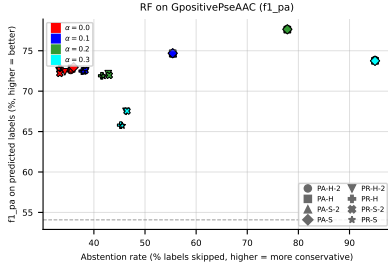


Figure 4: RF ($f1_{pa}$) on **GpositivePseAAC**: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

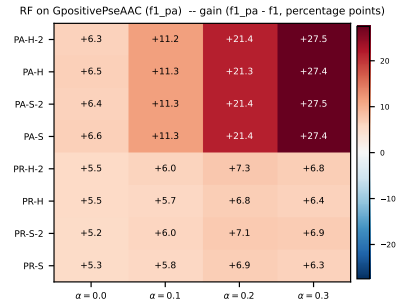
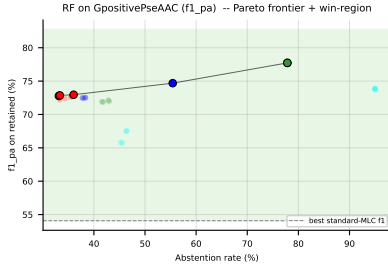


Figure 5: RF ($f1_{pa}$) on **GpositivePseAAC**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f_1$) heatmap (right).

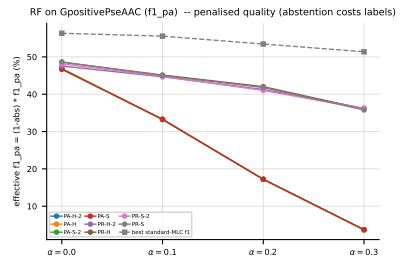
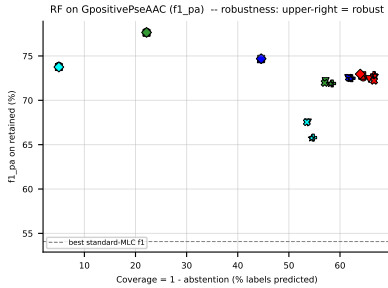


Figure 6: RF ($f1_{pa}$) on **GpositivePseAAC** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

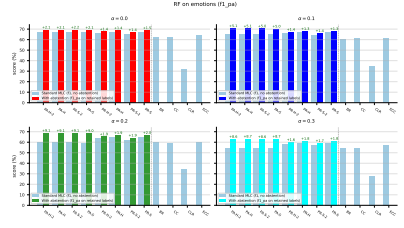
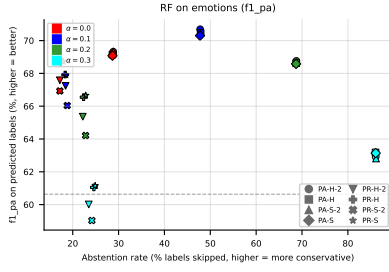


Figure 7: RF ($f1_{pa}$) on **emotions**: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

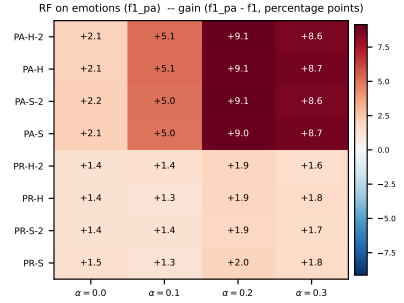
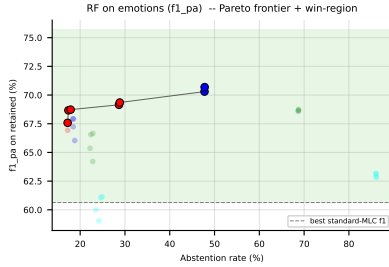


Figure 8: RF ($f1_{pa}$) on **emotions**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$, percentage points) heatmap (right).

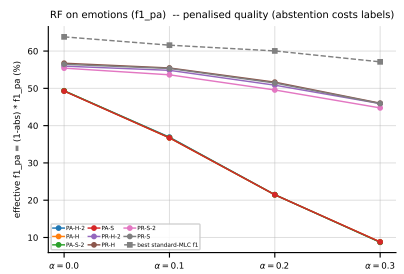
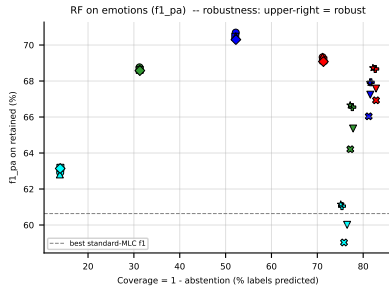


Figure 9: RF ($f1_{pa}$) on **emotions** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

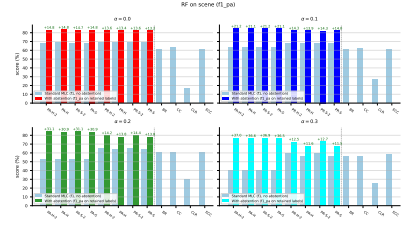
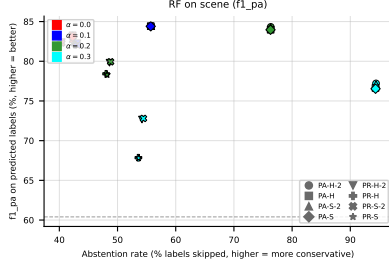


Figure 10: RF ($f1_{pa}$) on **scene**: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

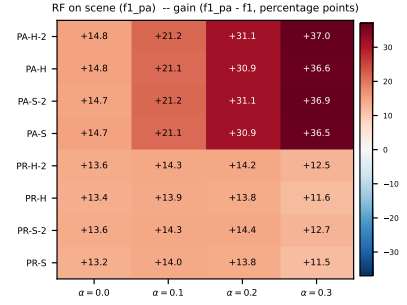
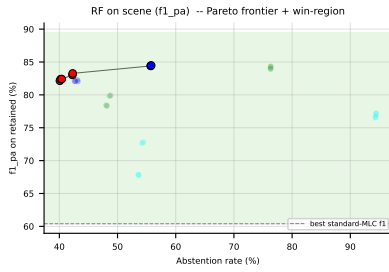


Figure 11: RF ($f1_{pa}$) on **scene**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

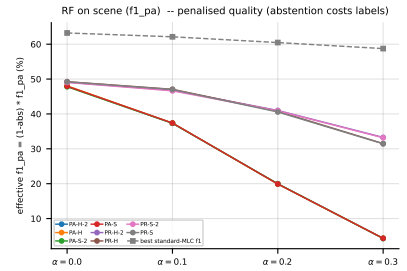
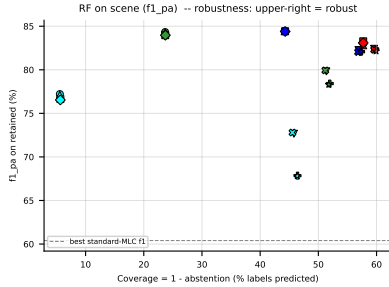


Figure 12: RF ($f1_{pa}$) on **scene** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

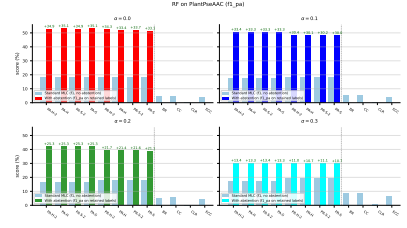
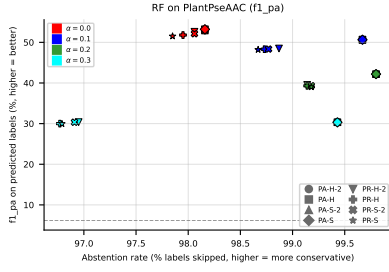


Figure 13: RF ($f1_{pa}$) on PlantPseAAC: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

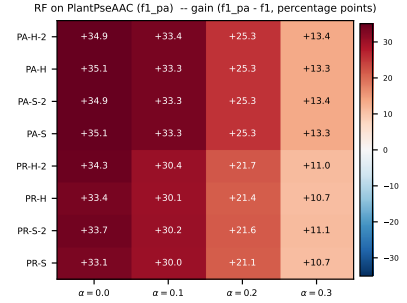
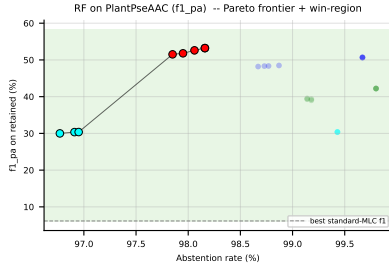


Figure 14: RF ($f1_{pa}$) on PlantPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f_1$) heatmap (right).

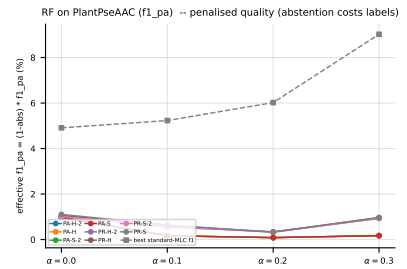
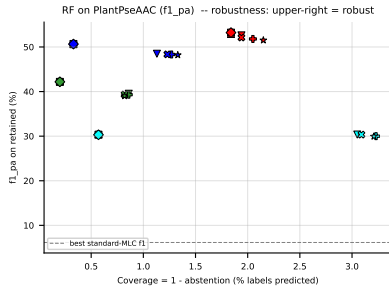


Figure 15: RF ($f1_{pa}$) on PlantPseAAC — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

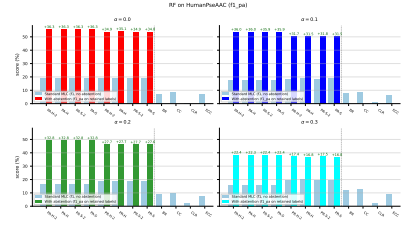
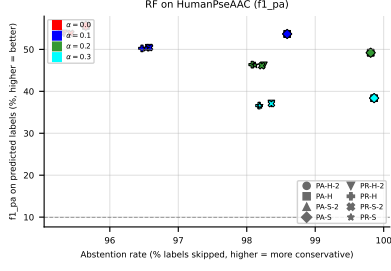


Figure 16: RF ($f1_{pa}$) on HumanPseAAC: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

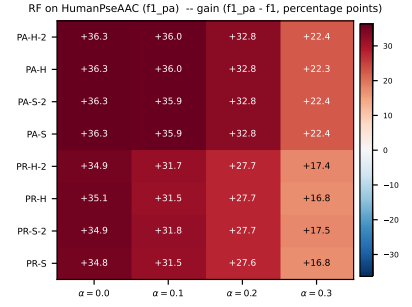
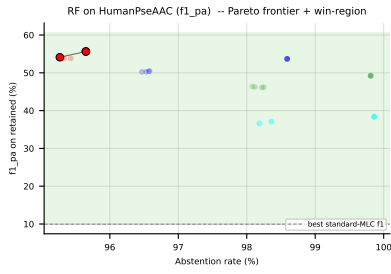


Figure 17: RF ($f1_{pa}$) on HumanPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

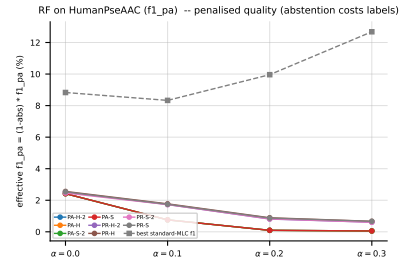
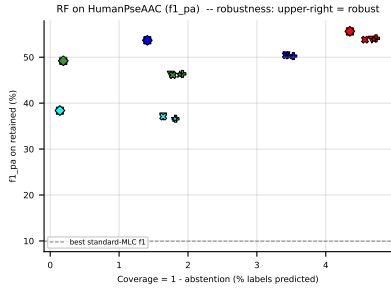


Figure 18: RF ($f1_{pa}$) on HumanPseAAC — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

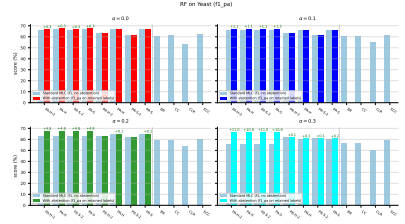
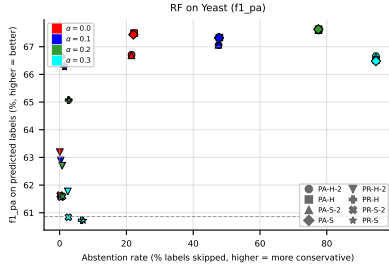


Figure 19: RF ($f1_{pa}$) on **Yeast**: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

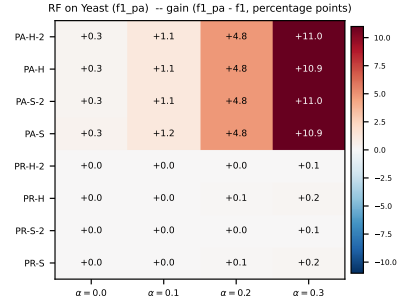
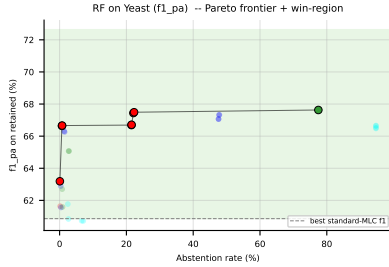


Figure 20: RF ($f1_{pa}$) on **Yeast**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

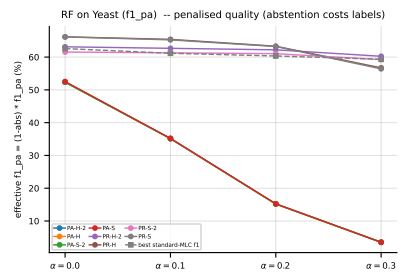
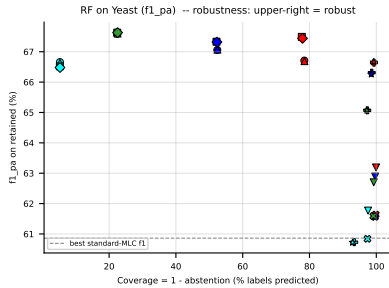


Figure 21: RF ($f1_{pa}$) on **Yeast** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

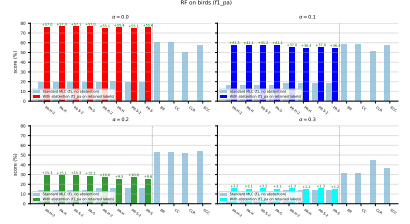
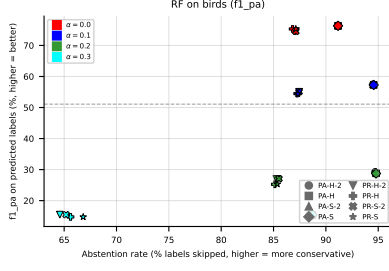


Figure 22: RF ($f1_{pa}$) on **birds**: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

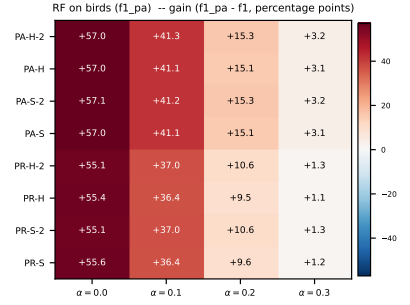
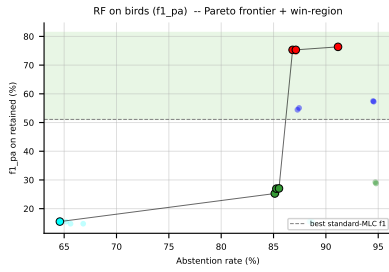


Figure 23: RF ($f1_{pa}$) on **birds**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

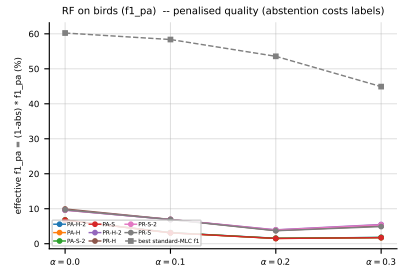
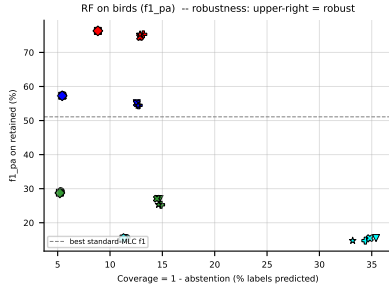


Figure 24: RF ($f1_{pa}$) on **birds** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

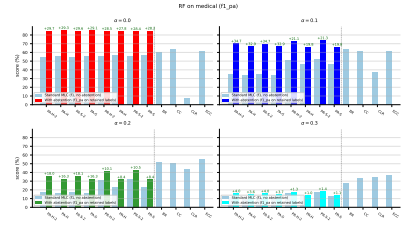
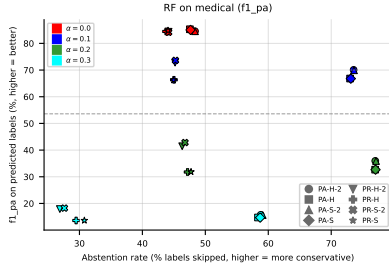


Figure 25: RF ($f1_{pa}$) on **medical**: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

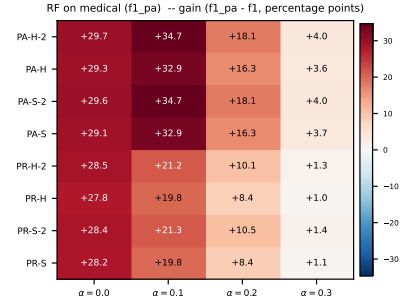
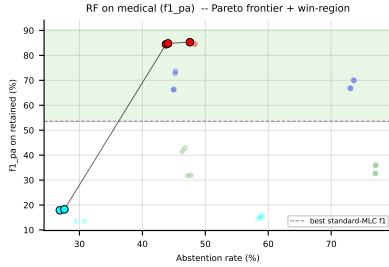


Figure 26: RF ($f1_{pa}$) on **medical**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

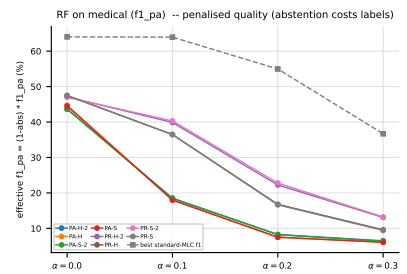
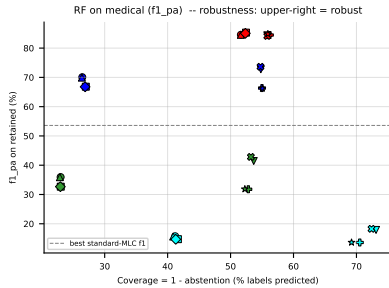


Figure 27: RF ($f1_{pa}$) on **medical** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

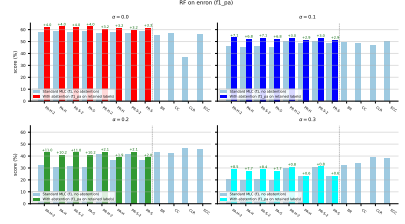
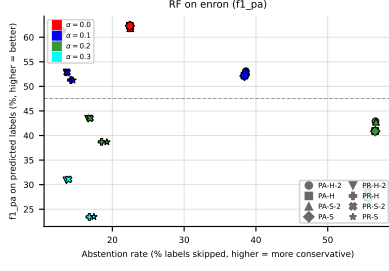


Figure 28: RF ($f1_pa$) on **enron**: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

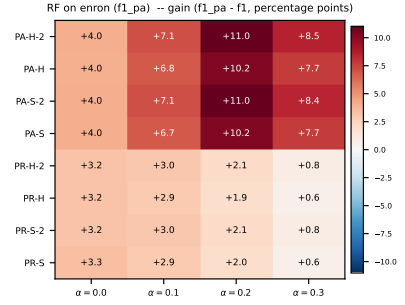
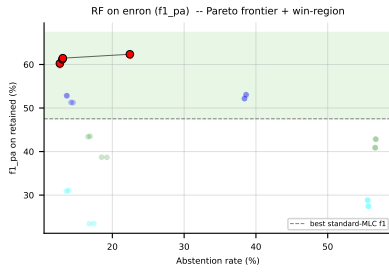


Figure 29: RF ($f1_pa$) on **enron**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_pa - f1$) heatmap (right).

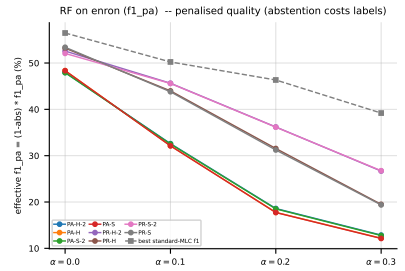
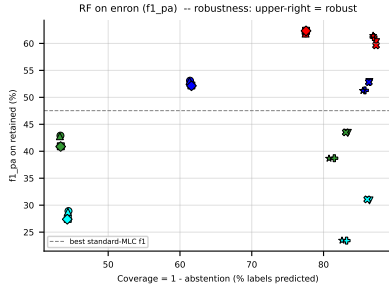


Figure 30: RF ($f1_pa$) on **enron** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_pa = (1 - \text{abs}) \cdot f1_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

4.3 RF, metric jaccard_pa — aggregate



Figure 31: RF (jaccard_pa) averaged across all datasets: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).



Figure 32: RF (jaccard_pa) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).



Figure 33: RF (jaccard_pa) averaged across all datasets — robustness view. Left: efficiency-quality scatter (x = coverage = 1 - abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective $\text{jaccard_pa} = (1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

4.4 RF, metric jaccard_pa — per dataset

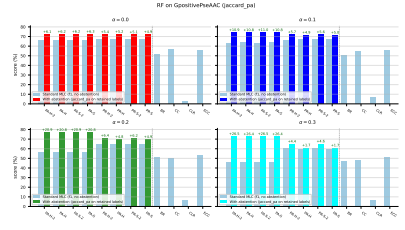
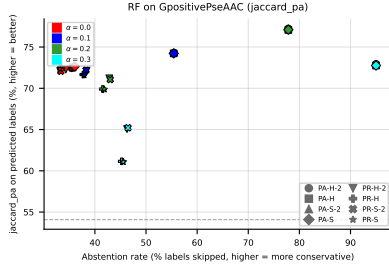


Figure 34: RF (jaccard_pa) on GpositivePseAAC: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

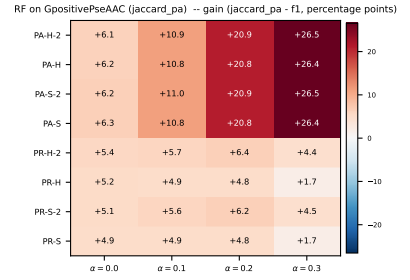
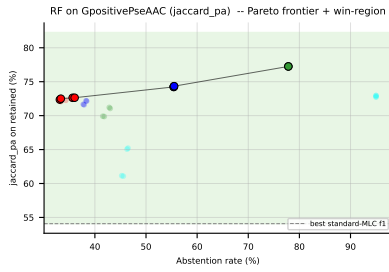


Figure 35: RF (jaccard_pa) on GpositivePseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

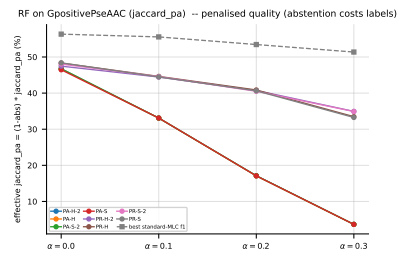
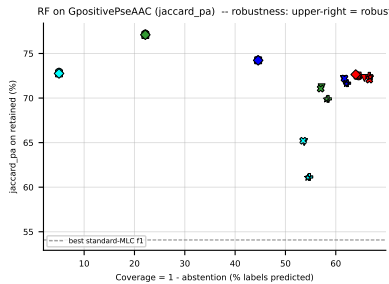


Figure 36: RF (jaccard_pa) on GpositivePseAAC — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $\text{jaccard_pa} = (1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

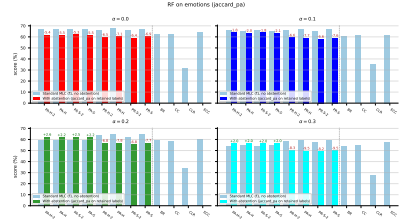
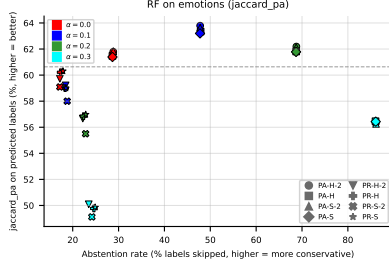


Figure 37: RF (jaccard_pa) on **emotions**: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

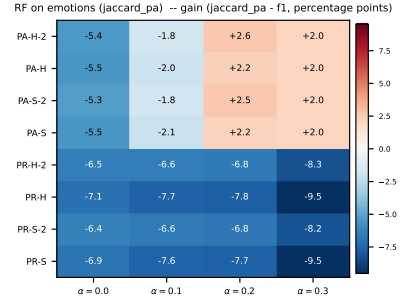
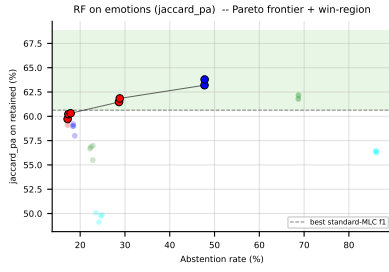


Figure 38: RF (jaccard_pa) on **emotions**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa - f_1) heatmap (right).

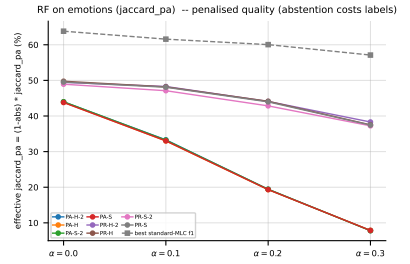
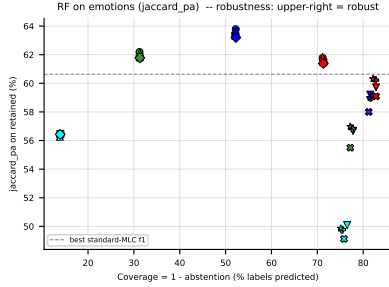


Figure 39: RF (jaccard_pa) on **emotions** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

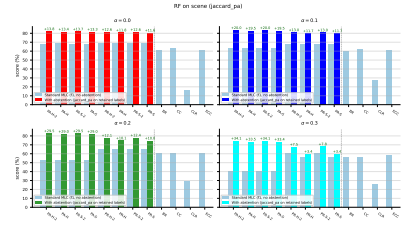
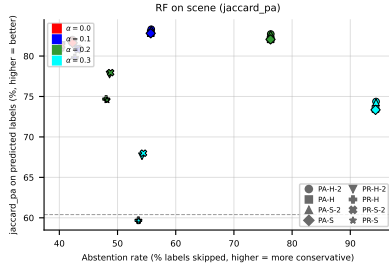


Figure 40: RF (jaccard_pa) on **scene**: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

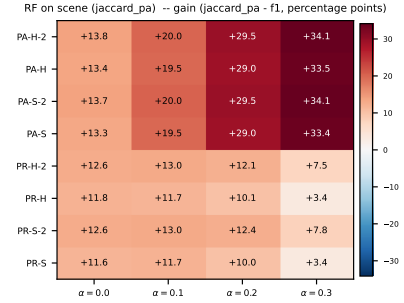
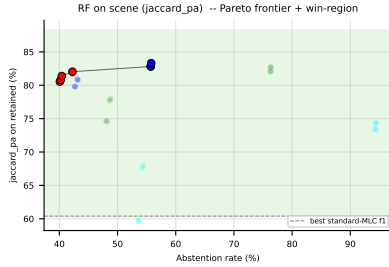


Figure 41: RF (jaccard_pa) on **scene**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa - f_1) heatmap (right).

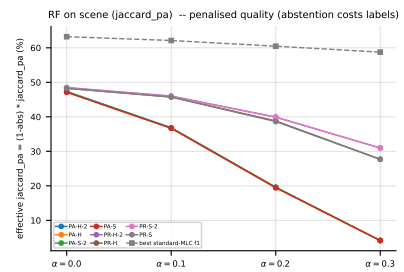
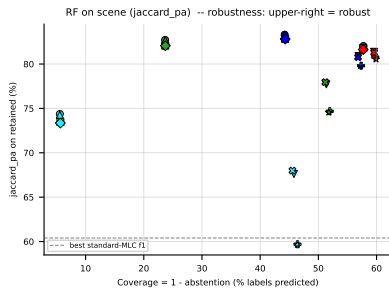


Figure 42: RF (jaccard_pa) on **scene** — robustness view. Left: efficiency-quality scatter (x = coverage = 1 - abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

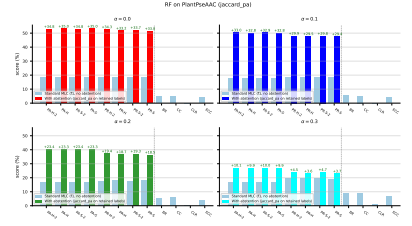
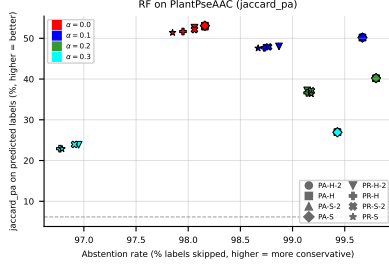


Figure 43: RF (jaccard_pa) on PlantPseAAC: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

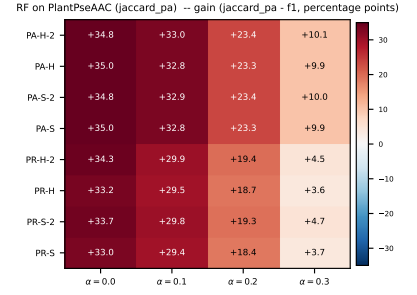
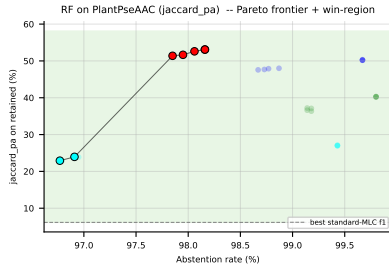


Figure 44: RF (jaccard_pa) on PlantPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($jaccard_pa - f_1$) heatmap (right).

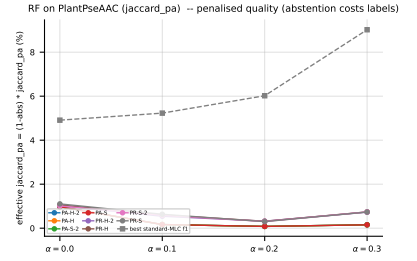
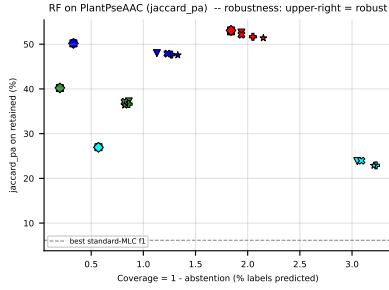


Figure 45: RF (jaccard_pa) on PlantPseAAC — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $jaccard_pa = (1 - \text{abs}) \cdot jaccard_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

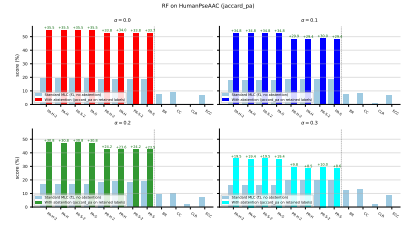
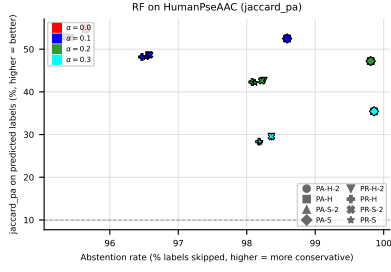


Figure 46: RF (jaccard_pa) on HumanPseAAC: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

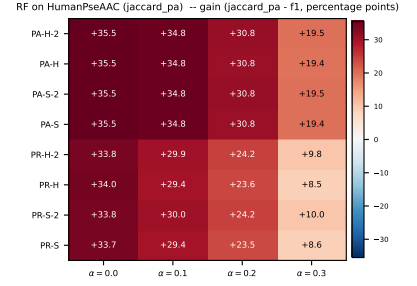
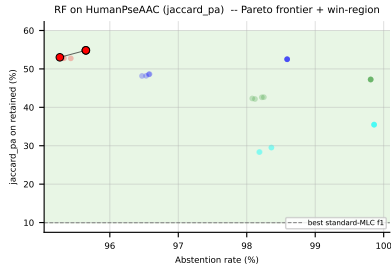


Figure 47: RF (jaccard_pa) on HumanPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$, percentage points) heatmap (right).

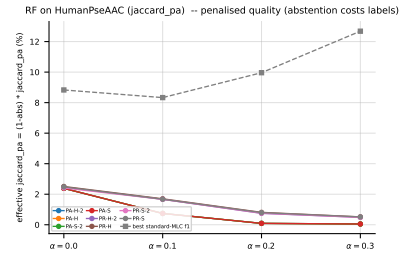
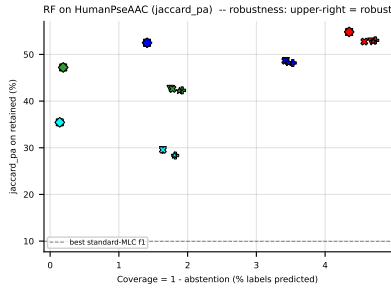


Figure 48: RF (jaccard_pa) on HumanPseAAC — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

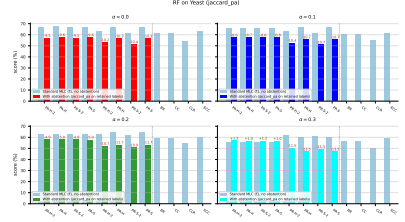
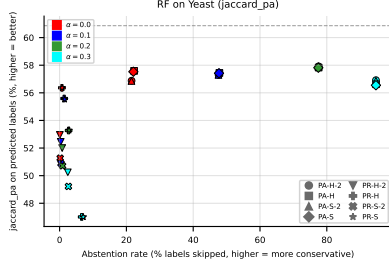


Figure 49: RF (jaccard_pa) on Yeast: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

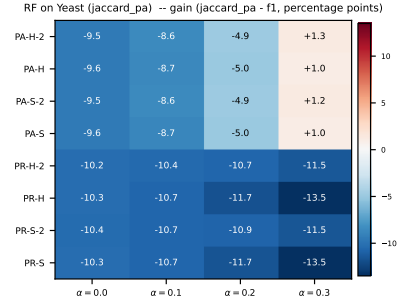
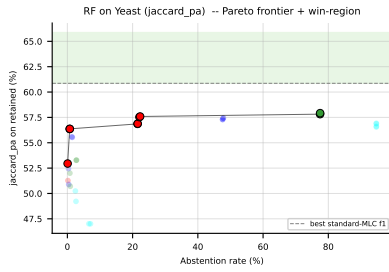


Figure 50: RF (jaccard_pa) on Yeast: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa - f_1) heatmap (right).

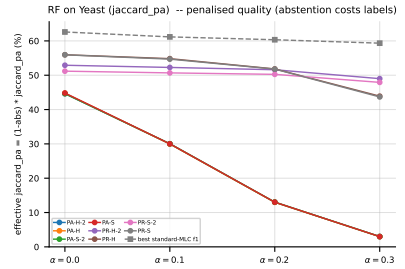
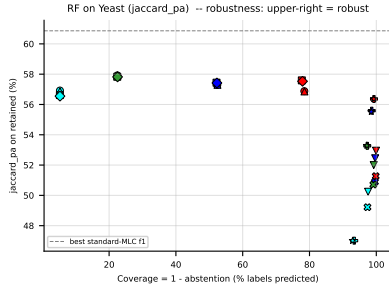


Figure 51: RF (jaccard_pa) on Yeast — robustness view. Left: efficiency-quality scatter (x = coverage = 1 - abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

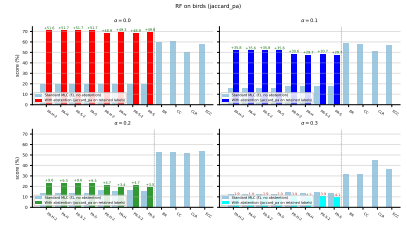
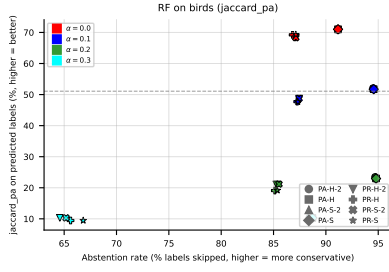


Figure 52: RF (jaccard_pa) on birds: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

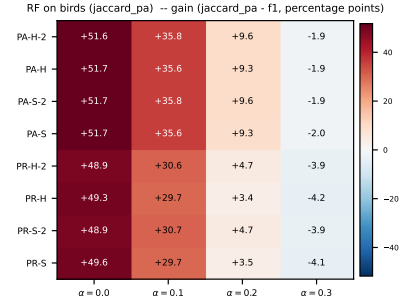
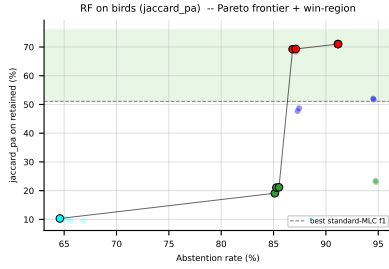


Figure 53: RF (jaccard_pa) on birds: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa - f_1) heatmap (right).

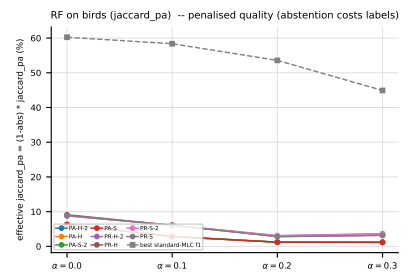
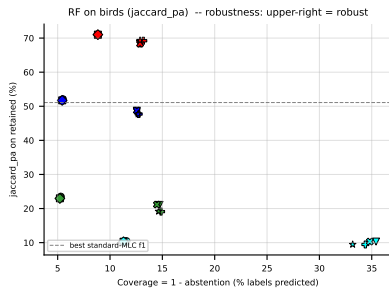


Figure 54: RF (jaccard_pa) on birds — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

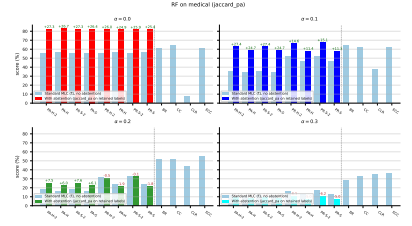
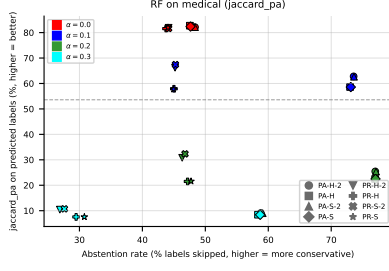


Figure 55: RF (jaccard_pa) on `medical`: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

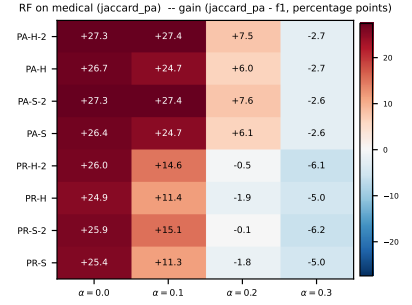
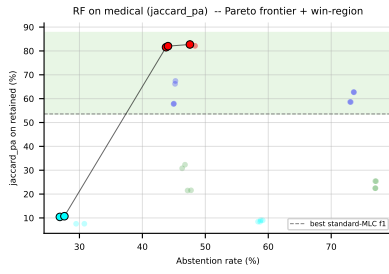


Figure 56: RF (jaccard_pa) on `medical`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

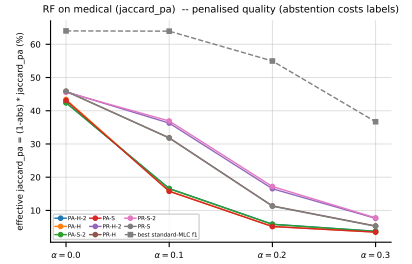
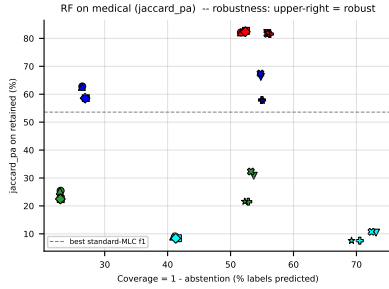


Figure 57: RF (jaccard_pa) on `medical` — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $\text{jaccard_pa} = (1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

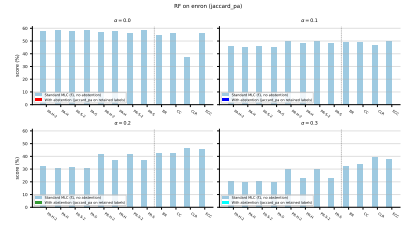
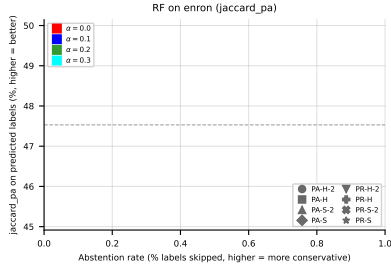


Figure 58: RF (jaccard_pa) on **enron**: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

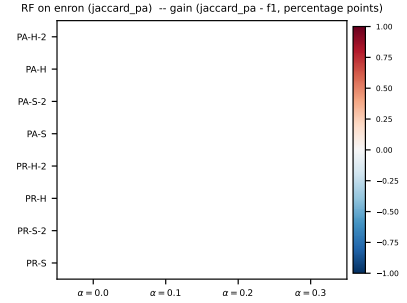
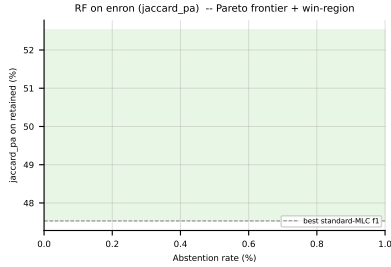


Figure 59: RF (jaccard_pa) on **enron**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

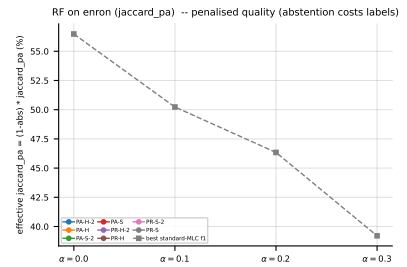
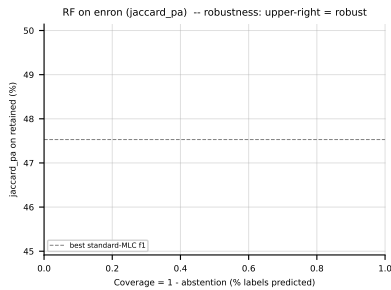


Figure 60: RF (jaccard_pa) on **enron** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $\text{jaccard_pa} = (1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

4.5 LGBM, metric $f1_{pa}$ — aggregate



Figure 61: LGBM ($f1_{pa}$) averaged across all datasets: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).



Figure 62: LGBM ($f1_{pa}$) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).



Figure 63: LGBM ($f1_{pa}$) averaged across all datasets — robustness view. Left: efficiency-quality scatter (x = coverage = 1 - abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - abs) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

4.6 LGBM, metric $f1_{pa}$ — per dataset

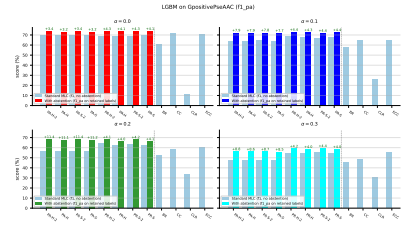
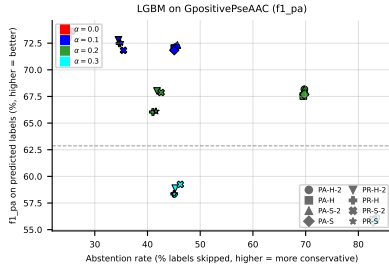


Figure 64: LGBM ($f1_pa$) on **GpositivePseAAC**: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

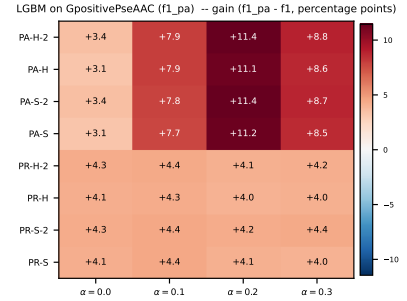
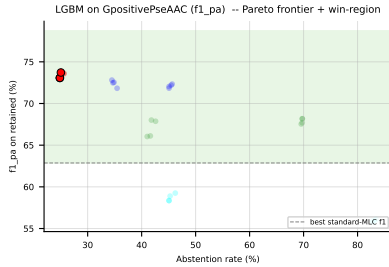


Figure 65: LGBM ($f1_pa$) on **GpositivePseAAC**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_pa - f1$) heatmap (right).

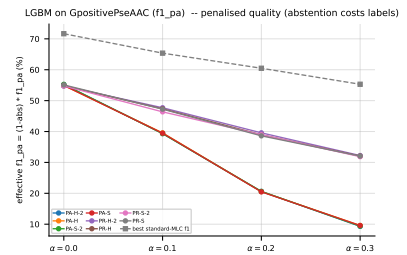
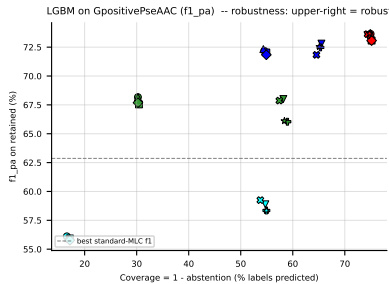


Figure 66: LGBM ($f1_pa$) on **GpositivePseAAC** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_pa = (1 - \text{abs}) \cdot f1_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

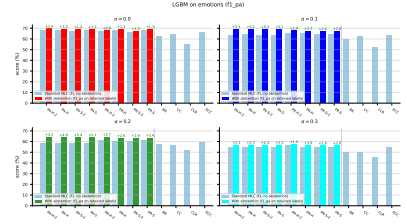
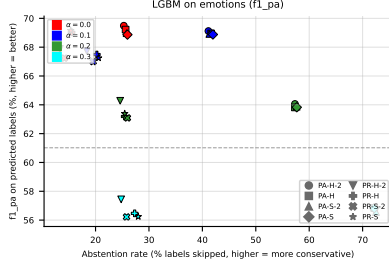


Figure 67: LGBM ($f1_{pa}$) on **emotions**: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

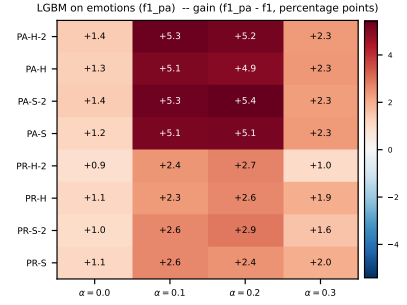
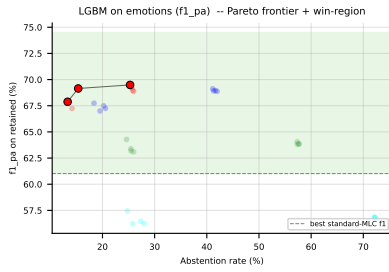


Figure 68: LGBM ($f1_{pa}$) on **emotions**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f_1$) heatmap (right).

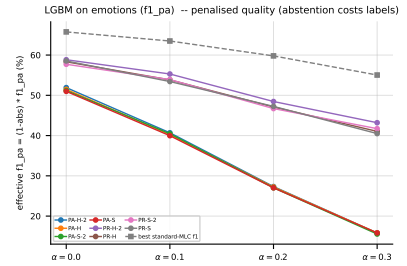
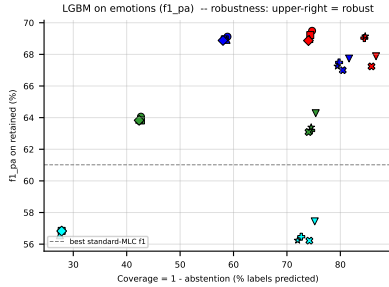


Figure 69: LGBM ($f1_{pa}$) on **emotions** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

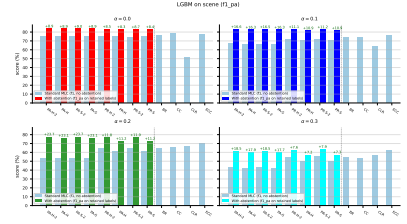
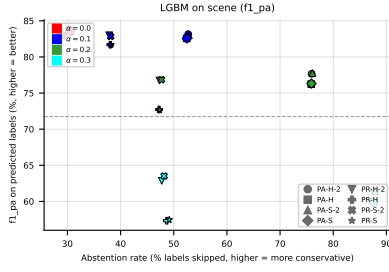


Figure 70: LGBM ($f1_{pa}$) on **scene**: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

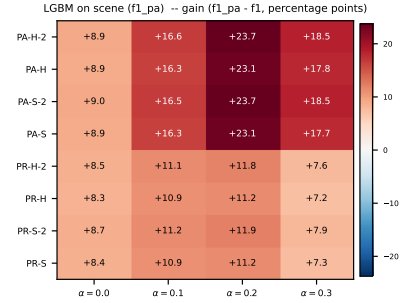
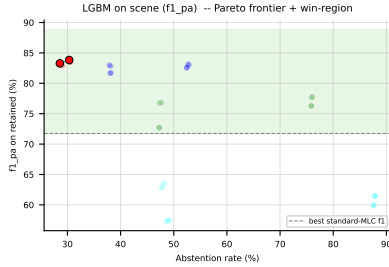


Figure 71: LGBM ($f1_{pa}$) on **scene**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

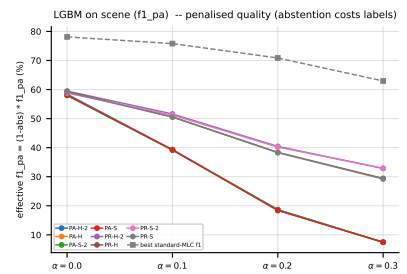
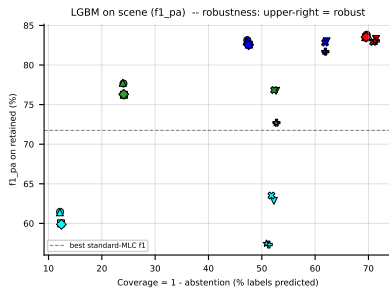


Figure 72: LGBM ($f1_{pa}$) on **scene** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

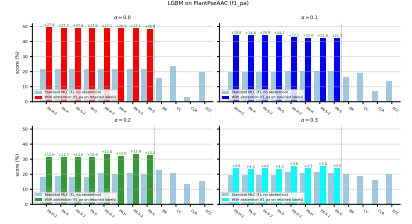
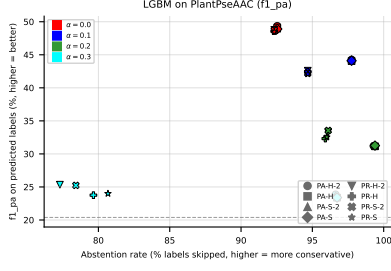


Figure 73: LGBM ($f1_{pa}$) on PlantPseAAC: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

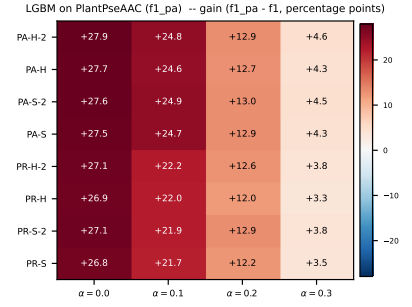
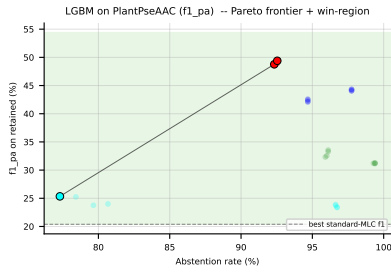


Figure 74: LGBM ($f1_{pa}$) on PlantPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f_1$) heatmap (right).

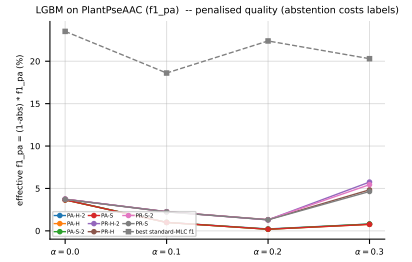
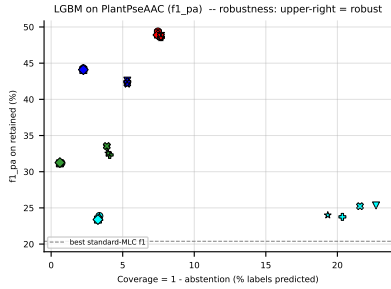


Figure 75: LGBM ($f1_{pa}$) on PlantPseAAC — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

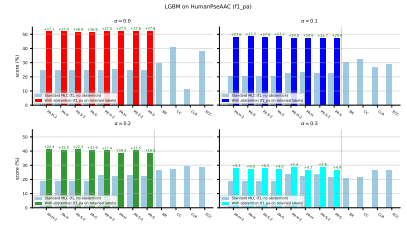
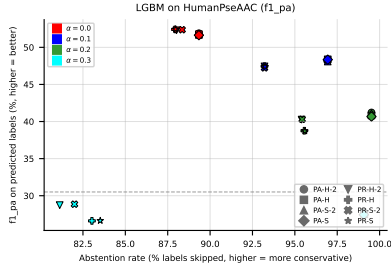


Figure 76: LGBM (f1_pa) on HumanPseAAC: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

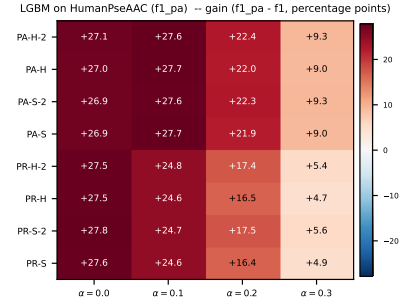
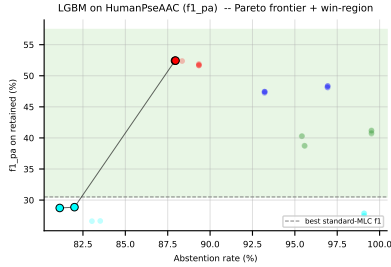


Figure 77: LGBM (f1_pa) on HumanPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f_1$) heatmap (right).

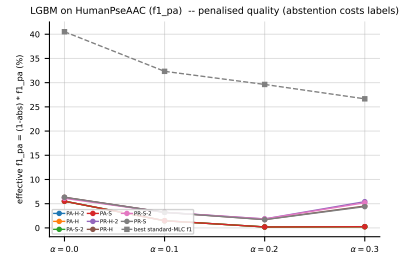
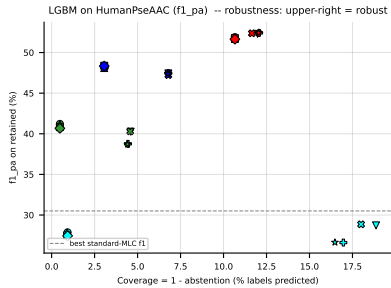


Figure 78: LGBM (f1_pa) on HumanPseAAC — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

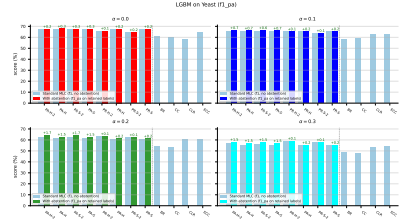
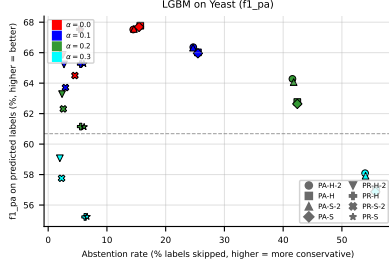


Figure 79: LGBM ($f1_pa$) on Yeast: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

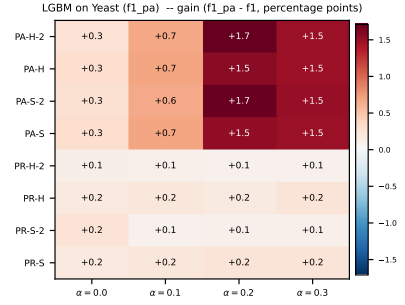
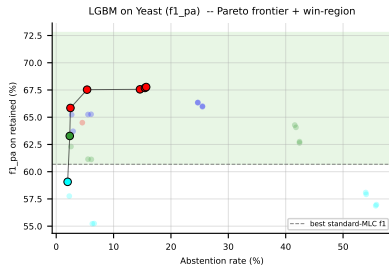


Figure 80: LGBM ($f1_pa$) on Yeast: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_pa - f1$) heatmap (right).

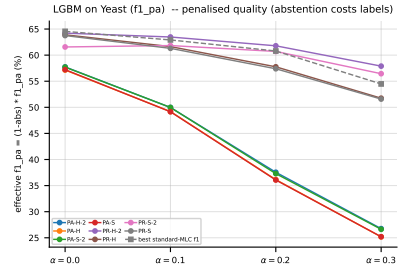
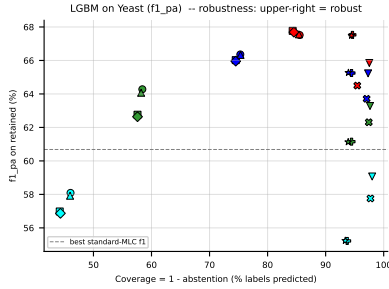


Figure 81: LGBM ($f1_pa$) on Yeast — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_pa = (1 - \text{abs}) \cdot f1_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

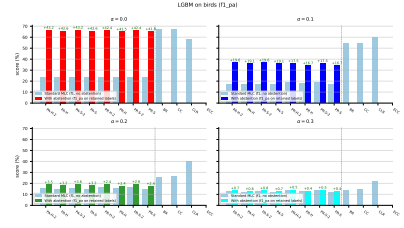
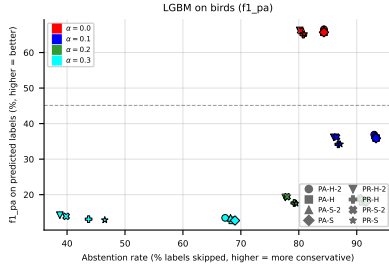


Figure 82: LGBM ($f1_{pa}$) on birds: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

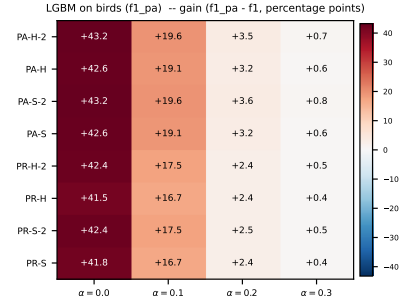
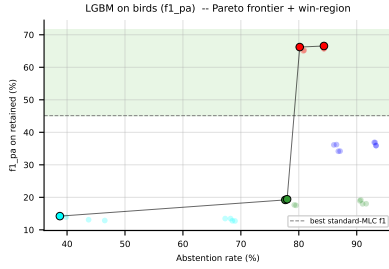


Figure 83: LGBM ($f1_{pa}$) on birds: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f_1$) heatmap (right).

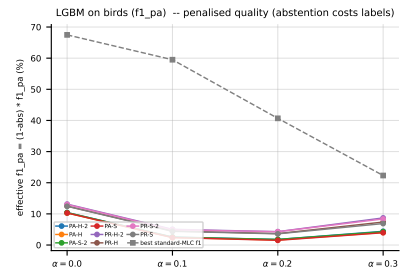
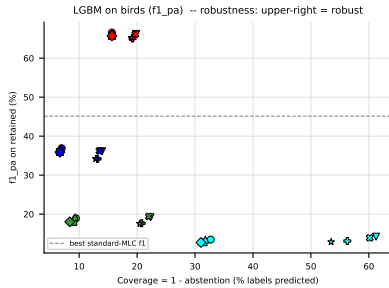


Figure 84: LGBM ($f1_{pa}$) on birds — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

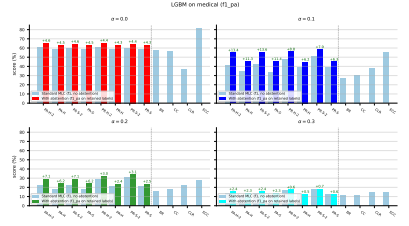
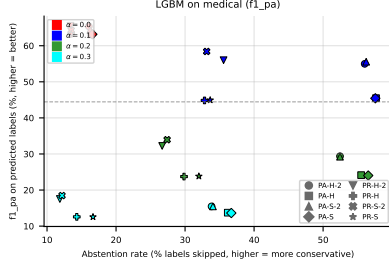


Figure 85: LGBM ($f1_pa$) on `medical`: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

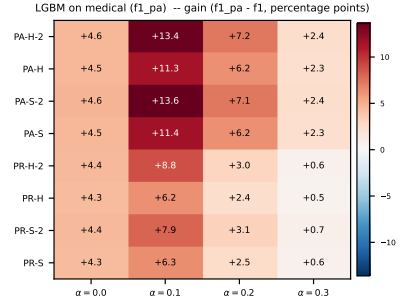
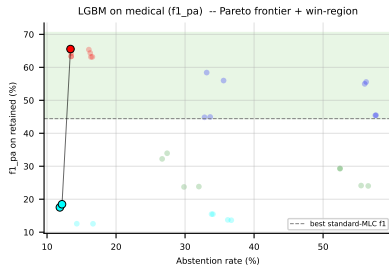


Figure 86: LGBM ($f1_pa$) on `medical`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_pa - f1$) heatmap (right).

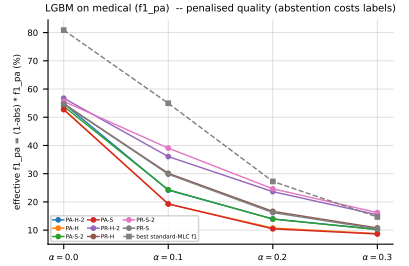
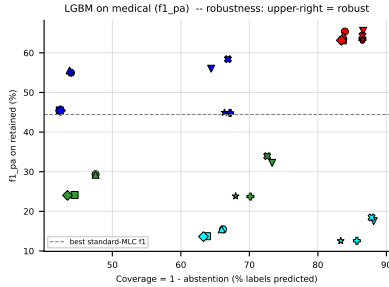


Figure 87: LGBM ($f1_pa$) on `medical` — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_pa = (1 - \text{abs}) \cdot f1_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

4.7 LGBM, metric jaccard_pa — aggregate



Figure 88: LGBM (jaccard_pa) averaged across all datasets: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).



Figure 89: LGBM (jaccard_pa) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).



Figure 90: LGBM (jaccard_pa) averaged across all datasets — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $\text{jaccard_pa} = (1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

4.8 LGBM, metric jaccard_pa — per dataset

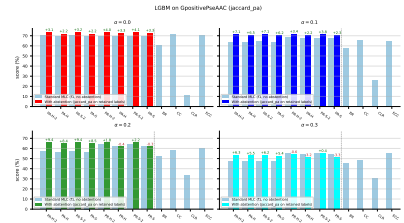
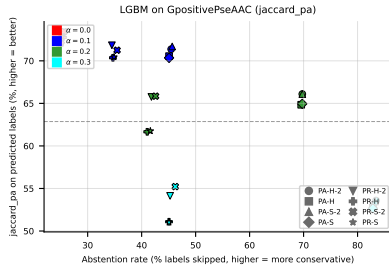


Figure 91: LGBM (jaccard_pa) on GpositivePseAAC: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

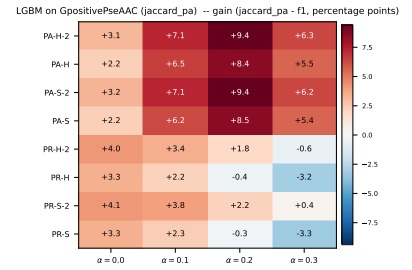
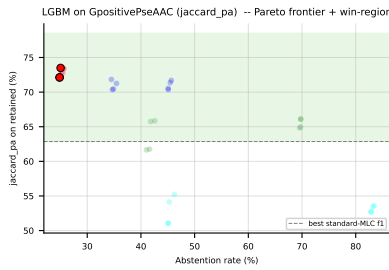


Figure 92: LGBM (jaccard_pa) on GpositivePseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa - f_1) heatmap (right).

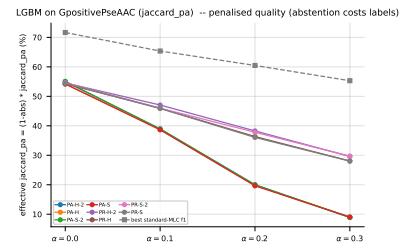
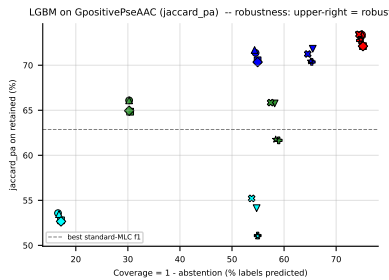


Figure 93: LGBM (jaccard_pa) on GpositivePseAAC — robustness view. Left: efficiency-quality scatter (x = coverage = 1 - abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

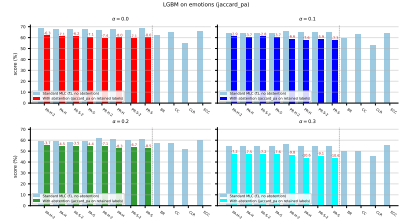
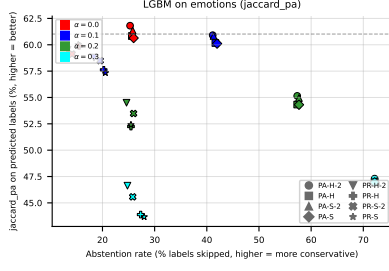


Figure 94: LGBM (jaccard_pa) on **emotions**: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

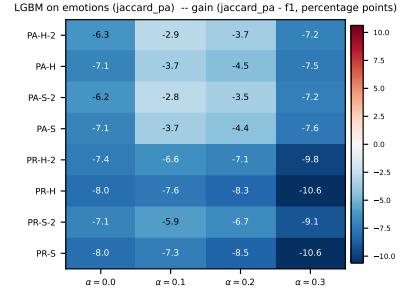
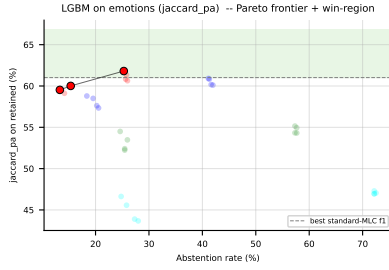


Figure 95: LGBM (jaccard_pa) on **emotions**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

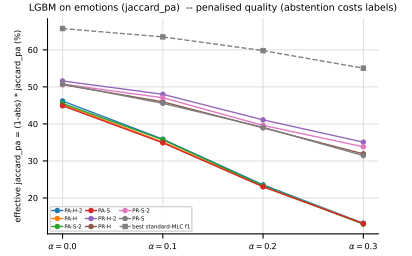
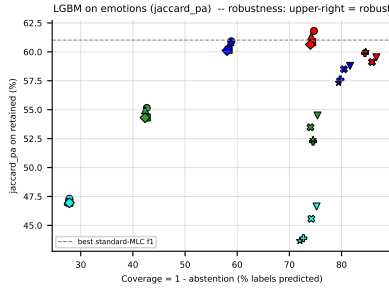


Figure 96: LGBM (jaccard_pa) on **emotions** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $\text{jaccard_pa} = (1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

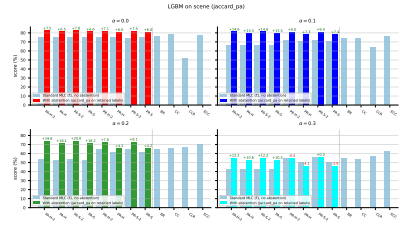
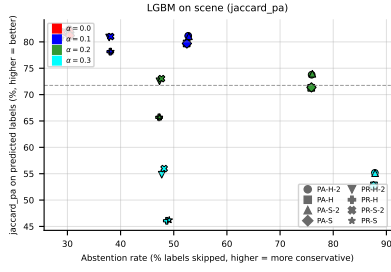


Figure 97: LGBM (jaccard_pa) on **scene**: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

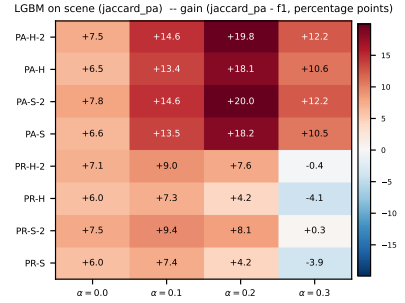
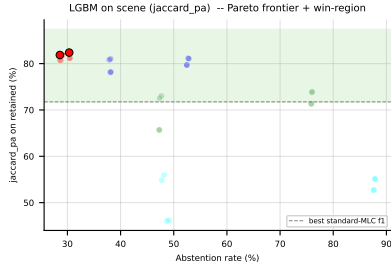


Figure 98: LGBM (jaccard_pa) on **scene**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa - f_1) heatmap (right).

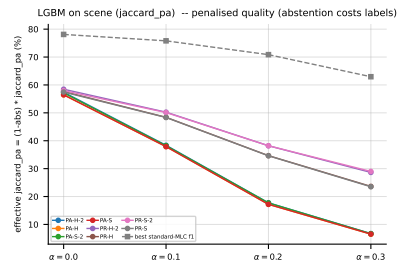
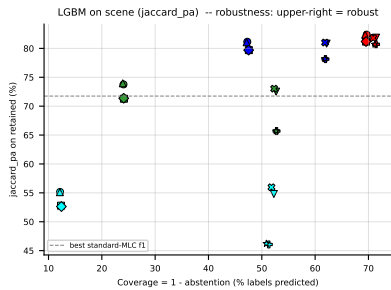


Figure 99: LGBM (jaccard_pa) on **scene** — robustness view. Left: efficiency-quality scatter (x = coverage = 1 - abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

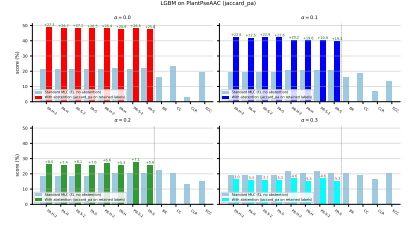
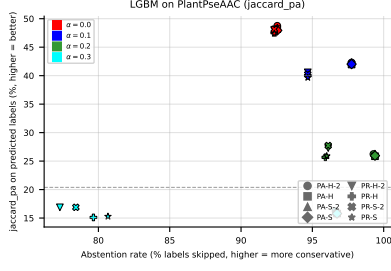


Figure 100: LGBM (jaccard_pa) on PlantPseAAC: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

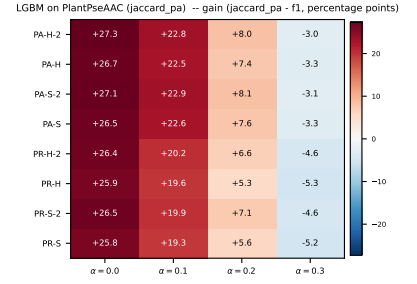
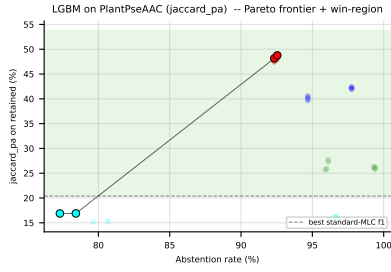


Figure 101: LGBM (jaccard_pa) on PlantPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

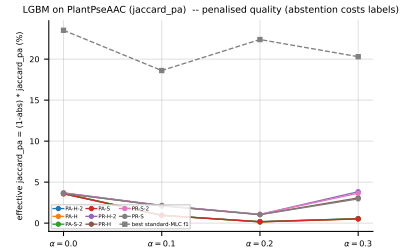
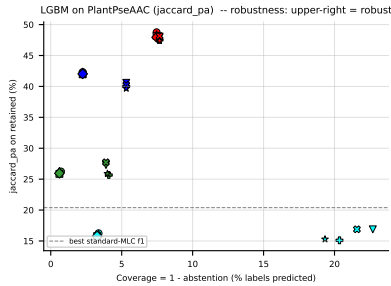


Figure 102: LGBM (jaccard_pa) on PlantPseAAC — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $\text{jaccard_pa} = (1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

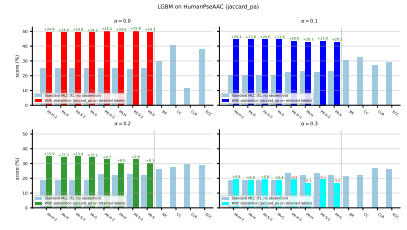
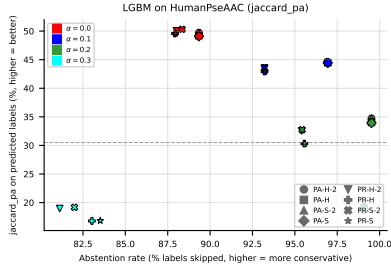


Figure 103: LGBM (jaccard_pa) on HumanPseAAC: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

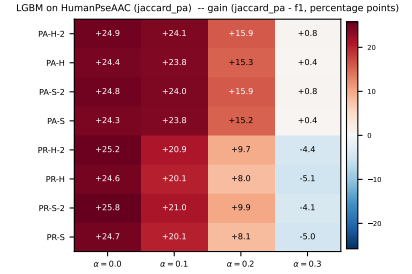
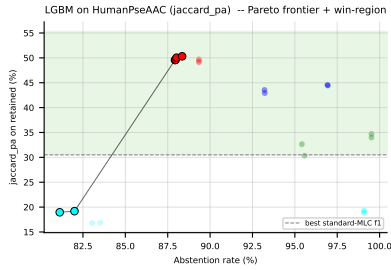


Figure 104: LGBM (jaccard_pa) on HumanPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa - f_1) heatmap (right).

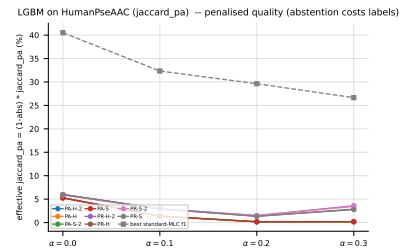
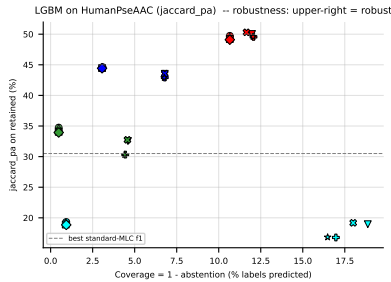


Figure 105: LGBM (jaccard_pa) on HumanPseAAC — robustness view. Left: efficiency-quality scatter (x = coverage = 1 - abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

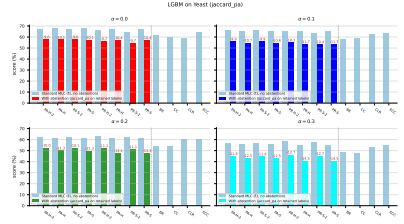
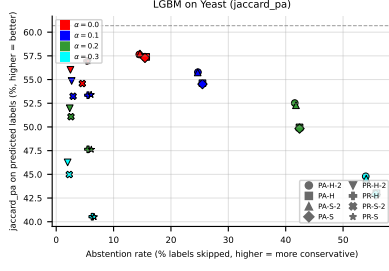


Figure 106: LGBM (jaccard_pa) on **Yeast**: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

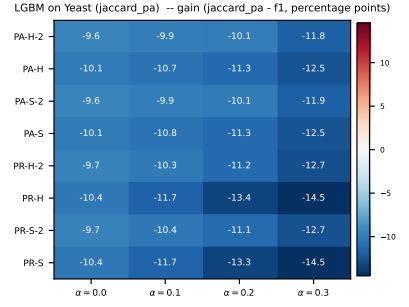
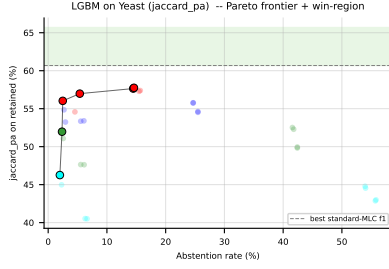


Figure 107: LGBM (jaccard_pa) on **Yeast**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa - f_1) heatmap (right).

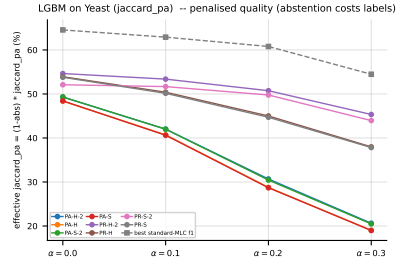
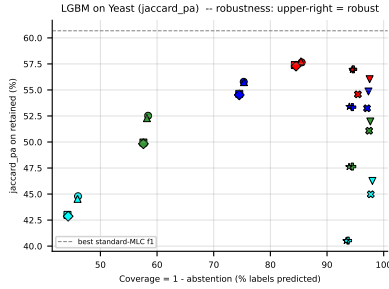


Figure 108: LGBM (jaccard_pa) on **Yeast** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

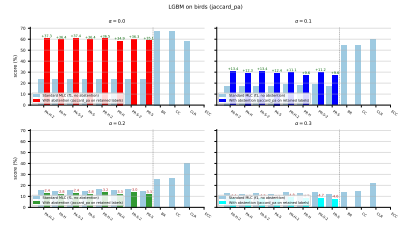
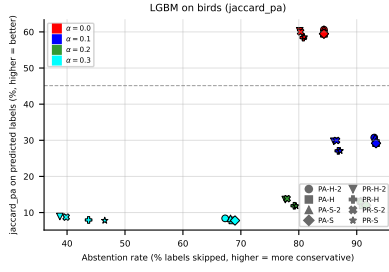


Figure 109: LGBM (jaccard_pa) on **birds**: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

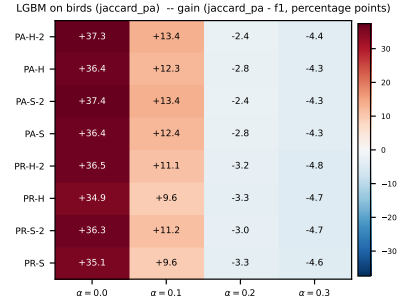
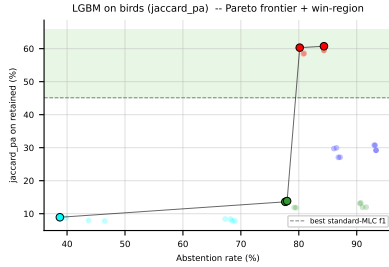


Figure 110: LGBM (jaccard_pa) on **birds**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($jaccard_pa - f_1$) heatmap (right).

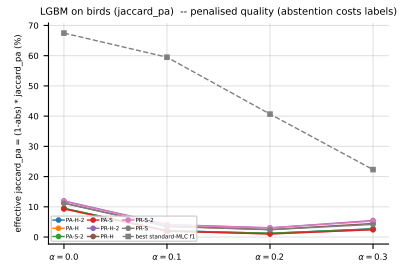
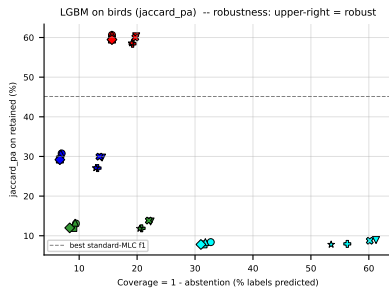


Figure 111: LGBM (jaccard_pa) on **birds** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $jaccard_pa = (1 - \text{abs}) \cdot jaccard_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

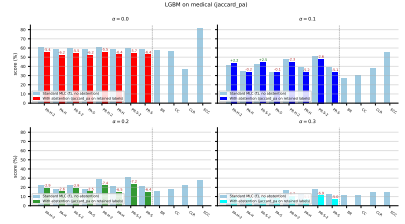
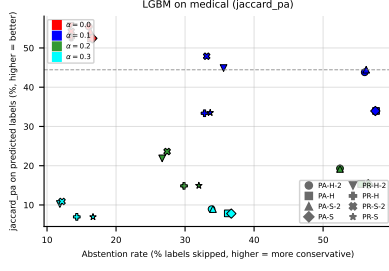


Figure 112: LGBM (jaccard_pa) on `medical`: coverage-risk scatter (left) and standard-MLC-vs-abstention paired bars across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$ (right).

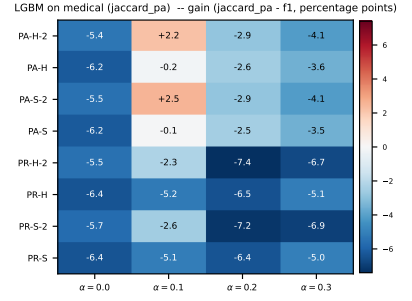
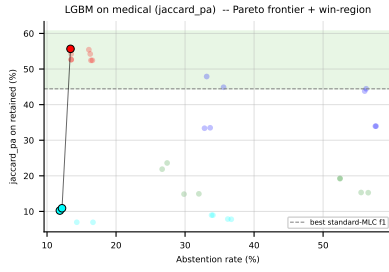


Figure 113: LGBM (jaccard_pa) on `medical`: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

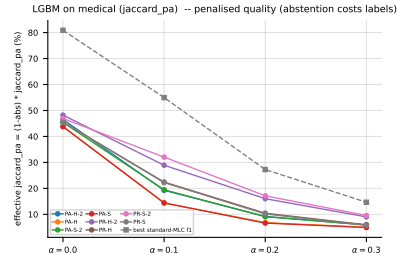
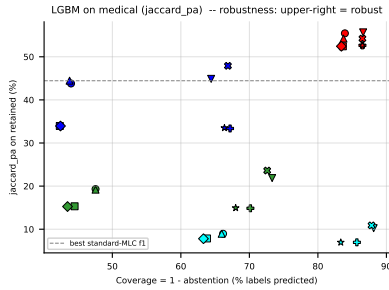


Figure 114: LGBM (jaccard_pa) on `medical` — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $\text{jaccard_pa} = (1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

A Per-dataset results (RF base learner)

A.1 GpositivePseAAC ($K = 4$)

RF • GpositivePseAAC • BinaryVector — Standard MLC predictions

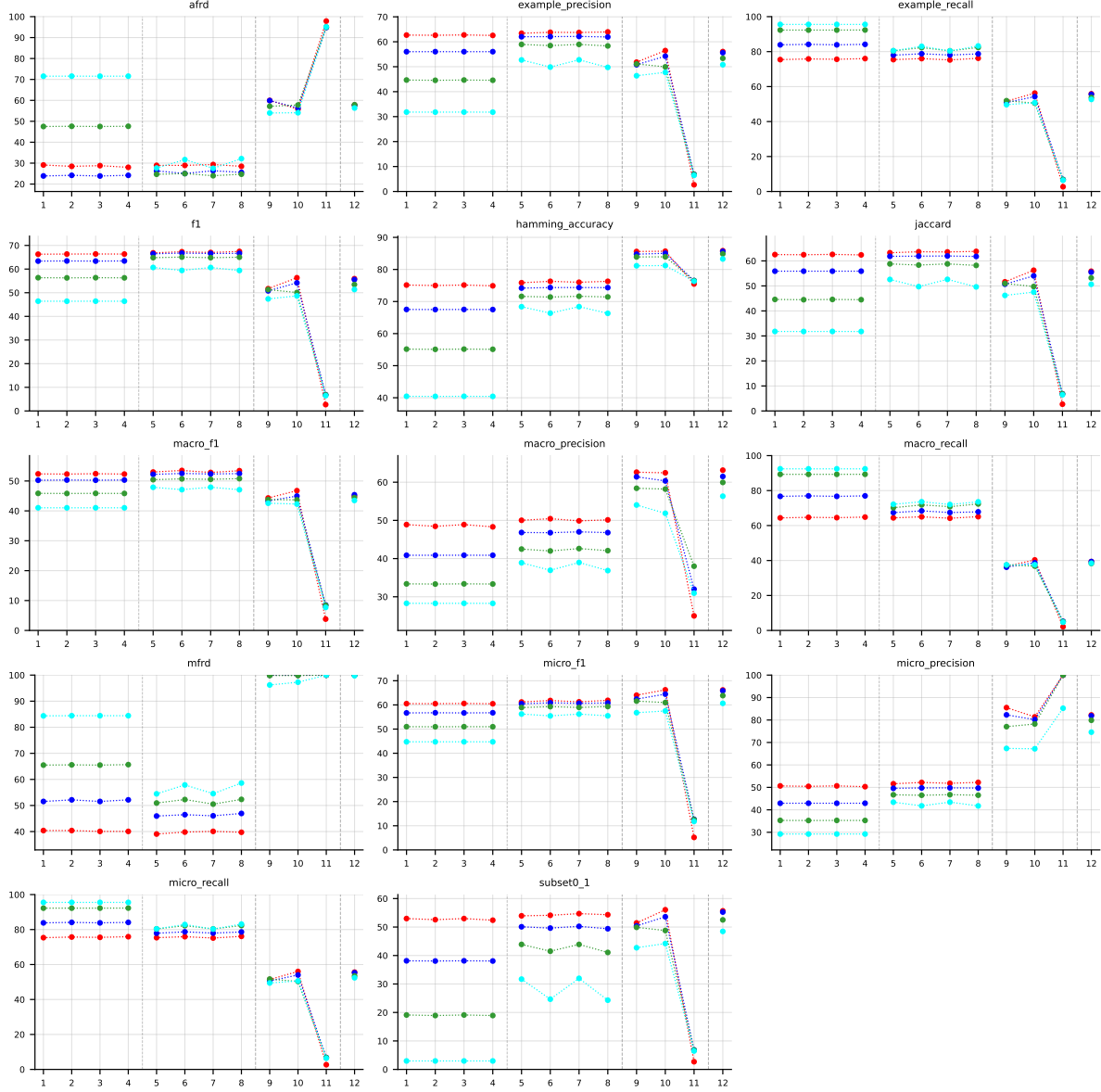


Table 5: Standard MLC predictions (BinaryVector) on GpositivePseAAC (RF base learner).

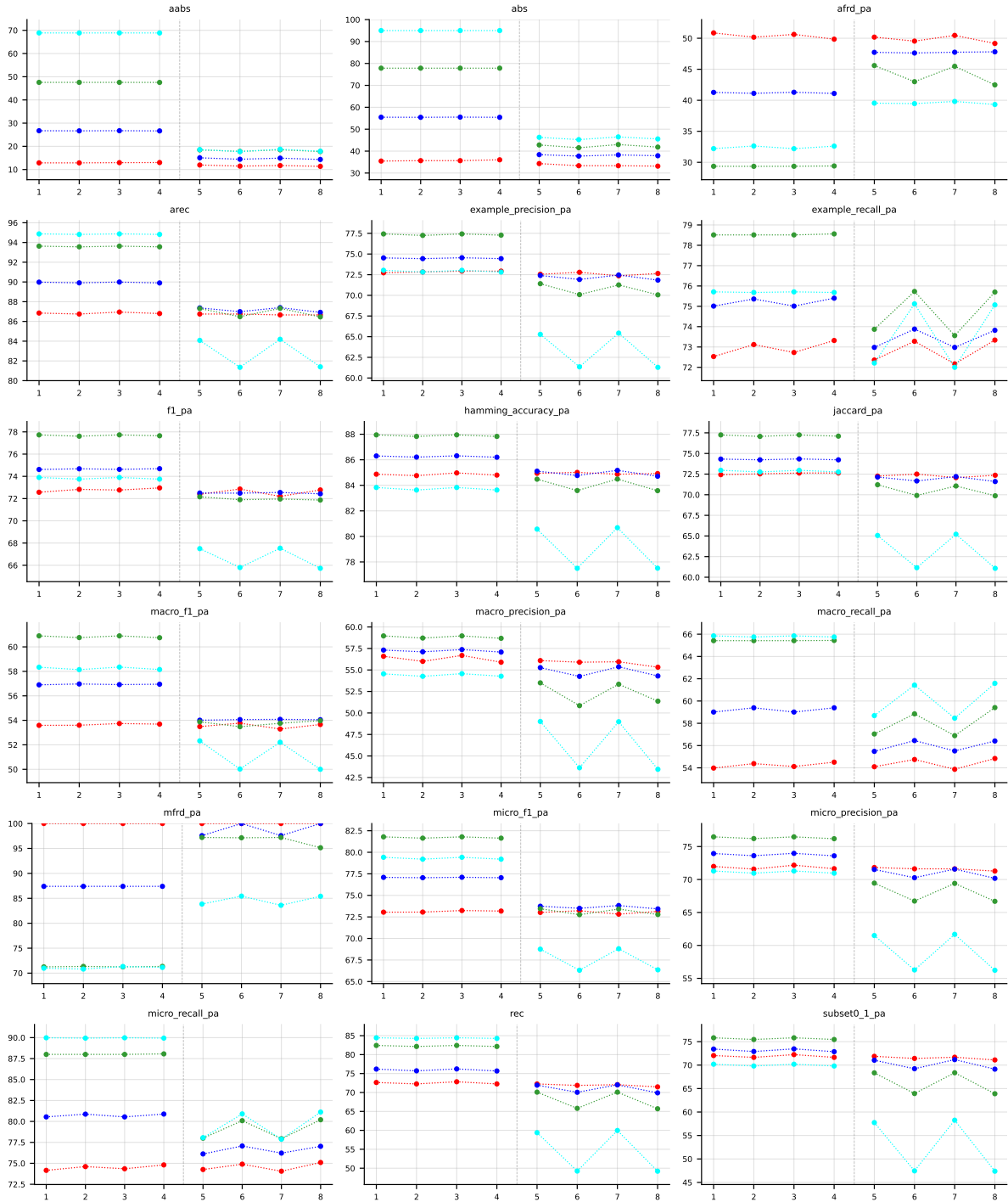


Table 6: Partial abstention (PartialAbstention) on GpositivePseAAC (RF base learner).

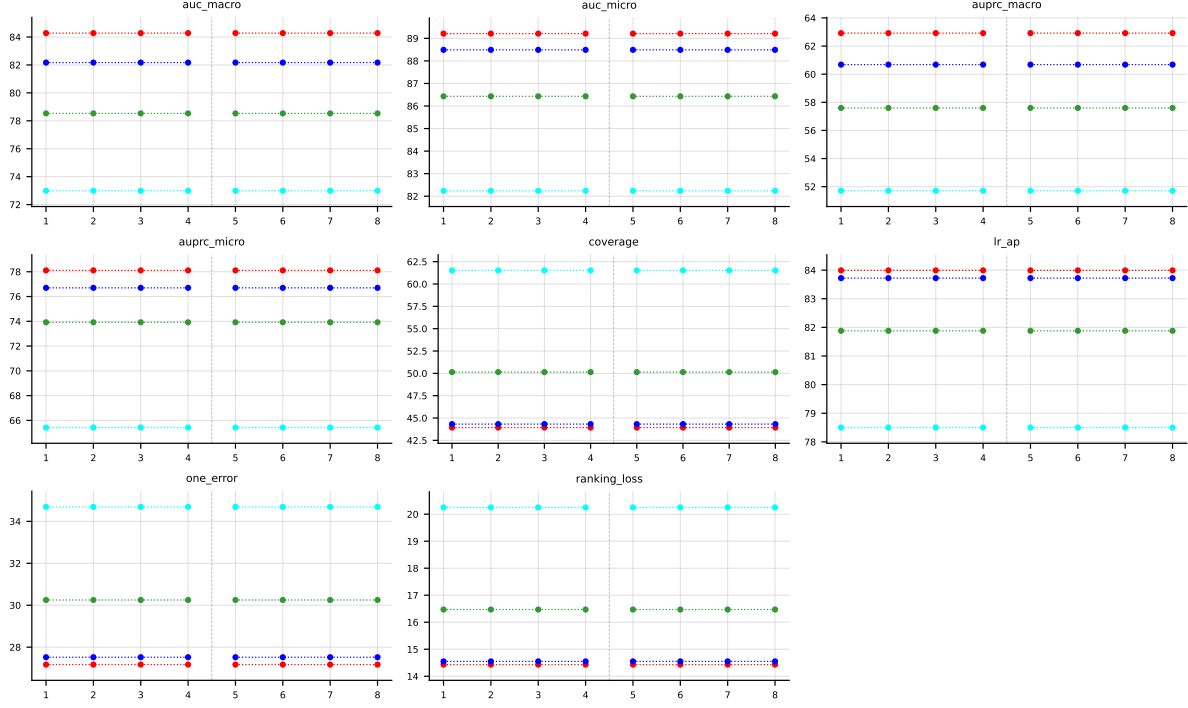


Table 7: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on GpositivePseAAC (RF base learner).

RF • GpositivePseAAC • PartialAbstention — Partial abstention

RF • GpositivePseAAC • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

A.2 Emotions ($K = 6$)

RF • emotions • BinaryVector — Standard MLC predictions

RF • emotions • PartialAbstention — Partial abstention

RF • emotions • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

A.3 Scene ($K = 6$)

RF • scene • BinaryVector — Standard MLC predictions

RF • scene • PartialAbstention — Partial abstention

RF • scene • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

A.4 PlantPseAAC ($K = 12$)

RF • PlantPseAAC • BinaryVector — Standard MLC predictions

RF • PlantPseAAC • PartialAbstention — Partial abstention

RF • PlantPseAAC • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

A.5 HumanPseAAC ($K = 14$)

RF • HumanPseAAC • BinaryVector — Standard MLC predictions

RF • HumanPseAAC • PartialAbstention — Partial abstention

RF • HumanPseAAC • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

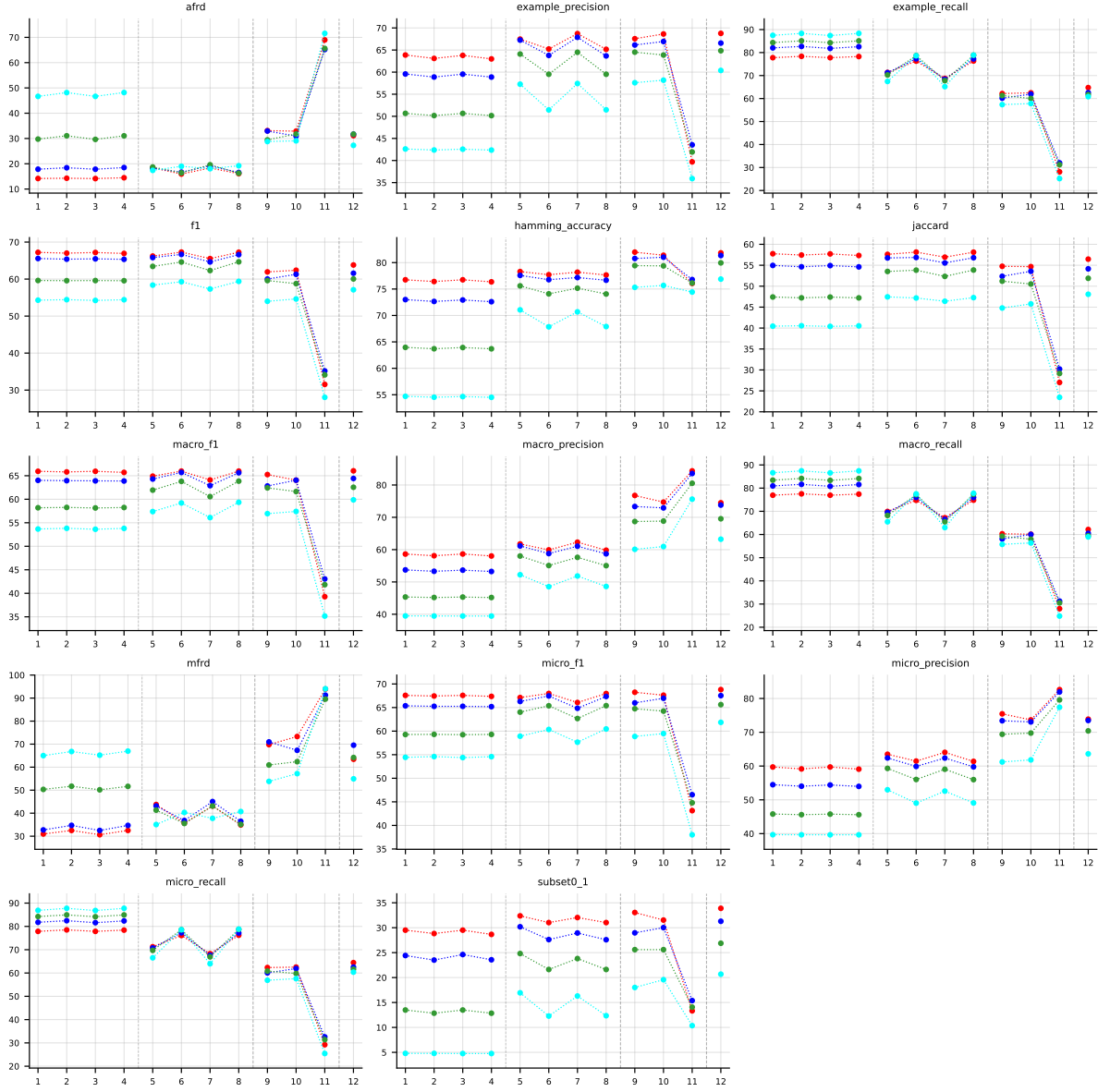


Table 8: Standard MLC predictions (BinaryVector) on `emotions` (RF base learner).

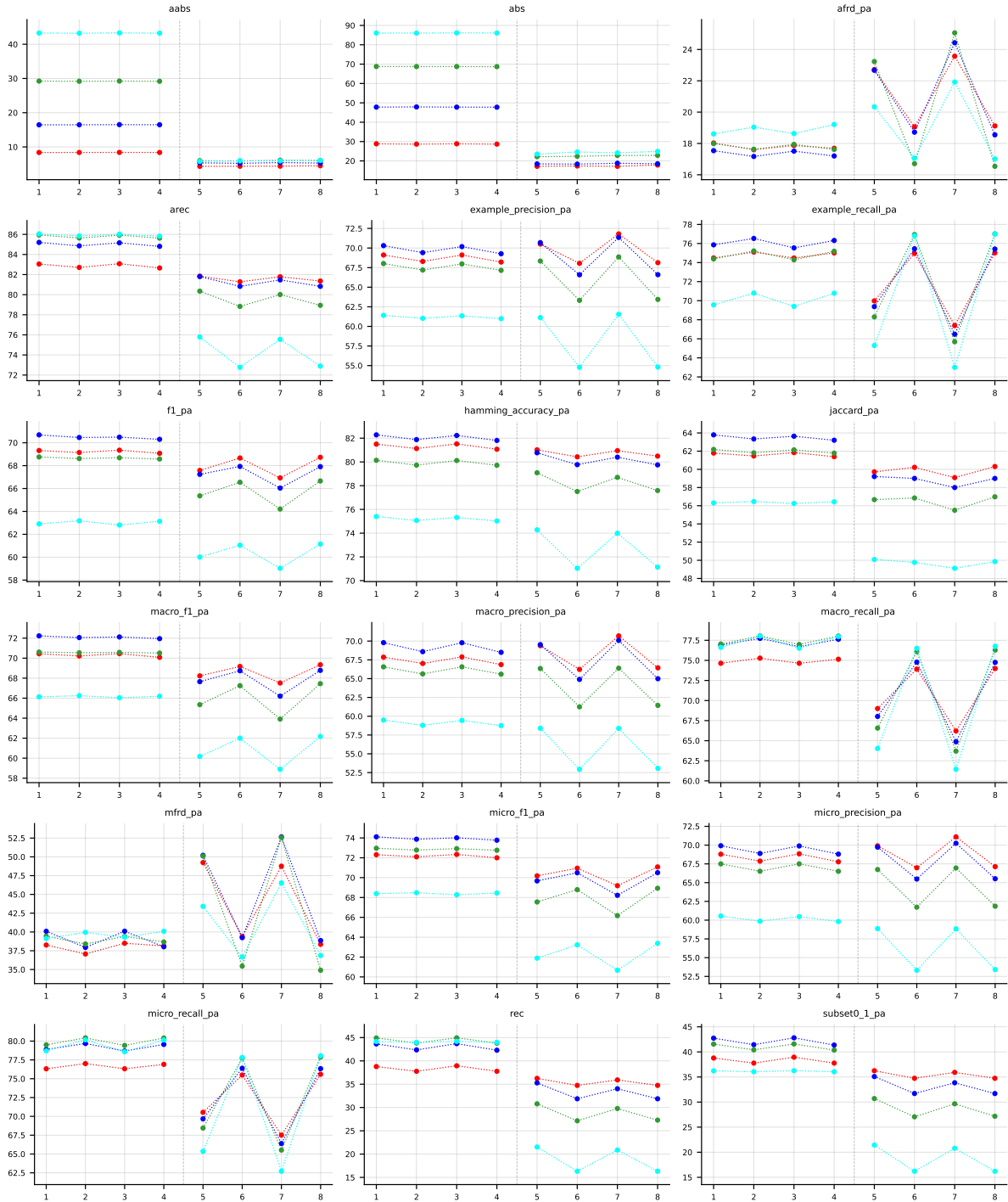


Table 9: Partial abstention (PartialAbstention) on emotions (RF base learner).

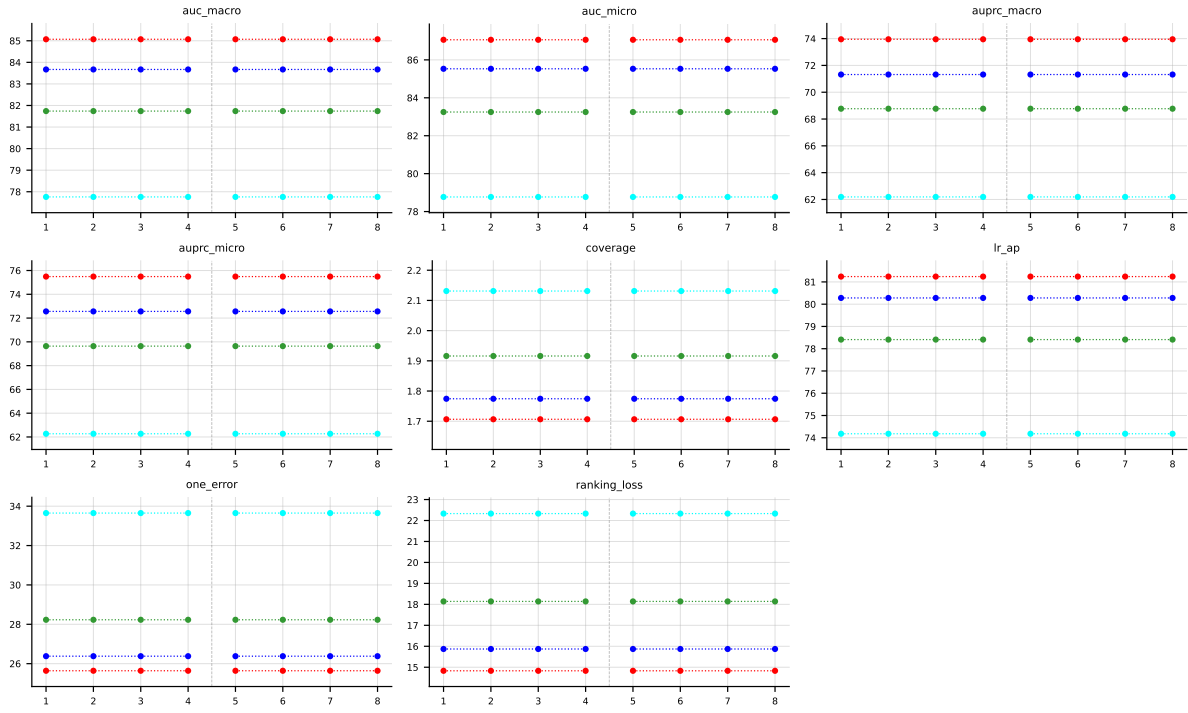


Table 10: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on **emotions** (RF base learner).

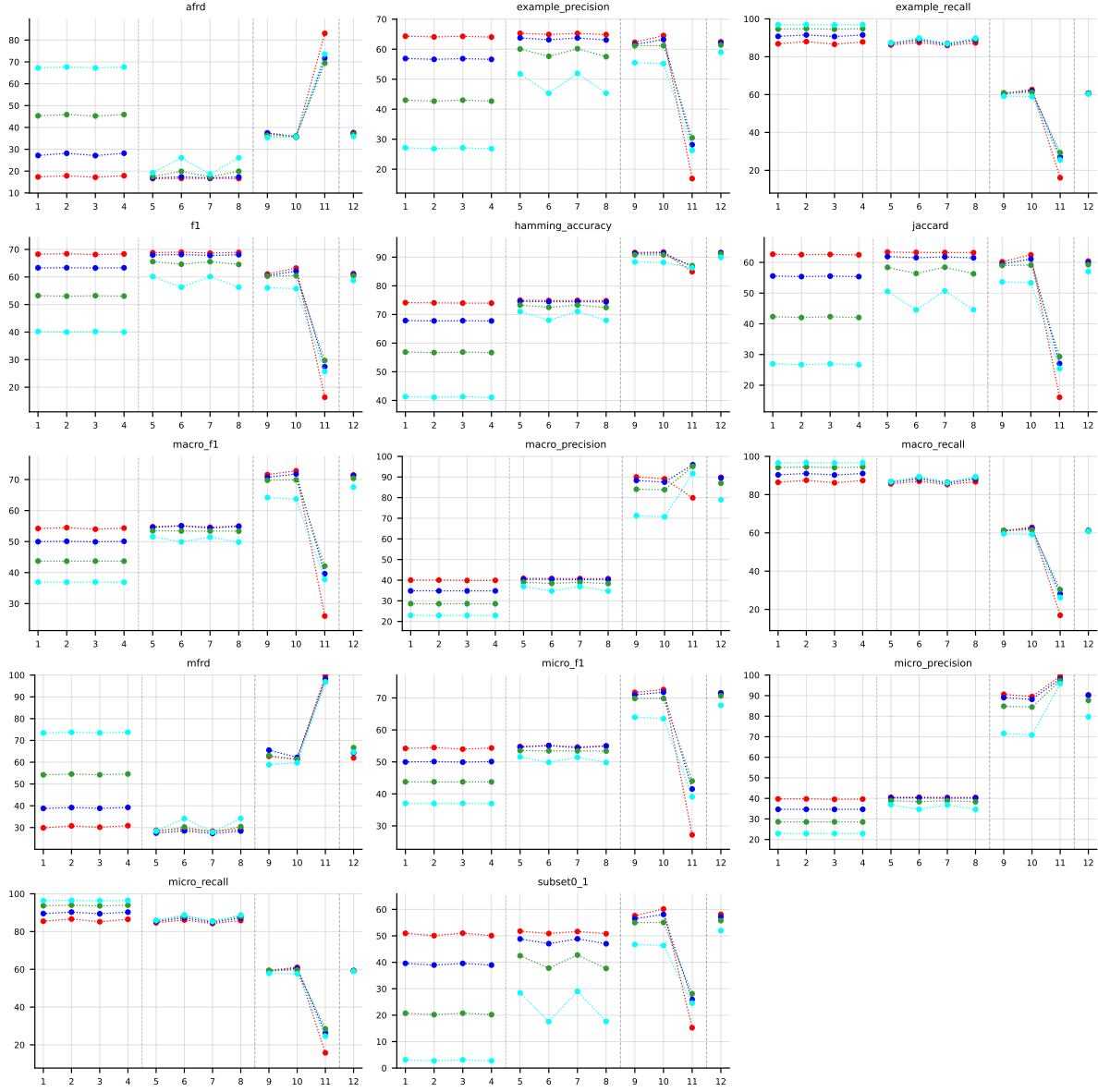


Table 11: Standard MLC predictions (BinaryVector) on scene (RF base learner).

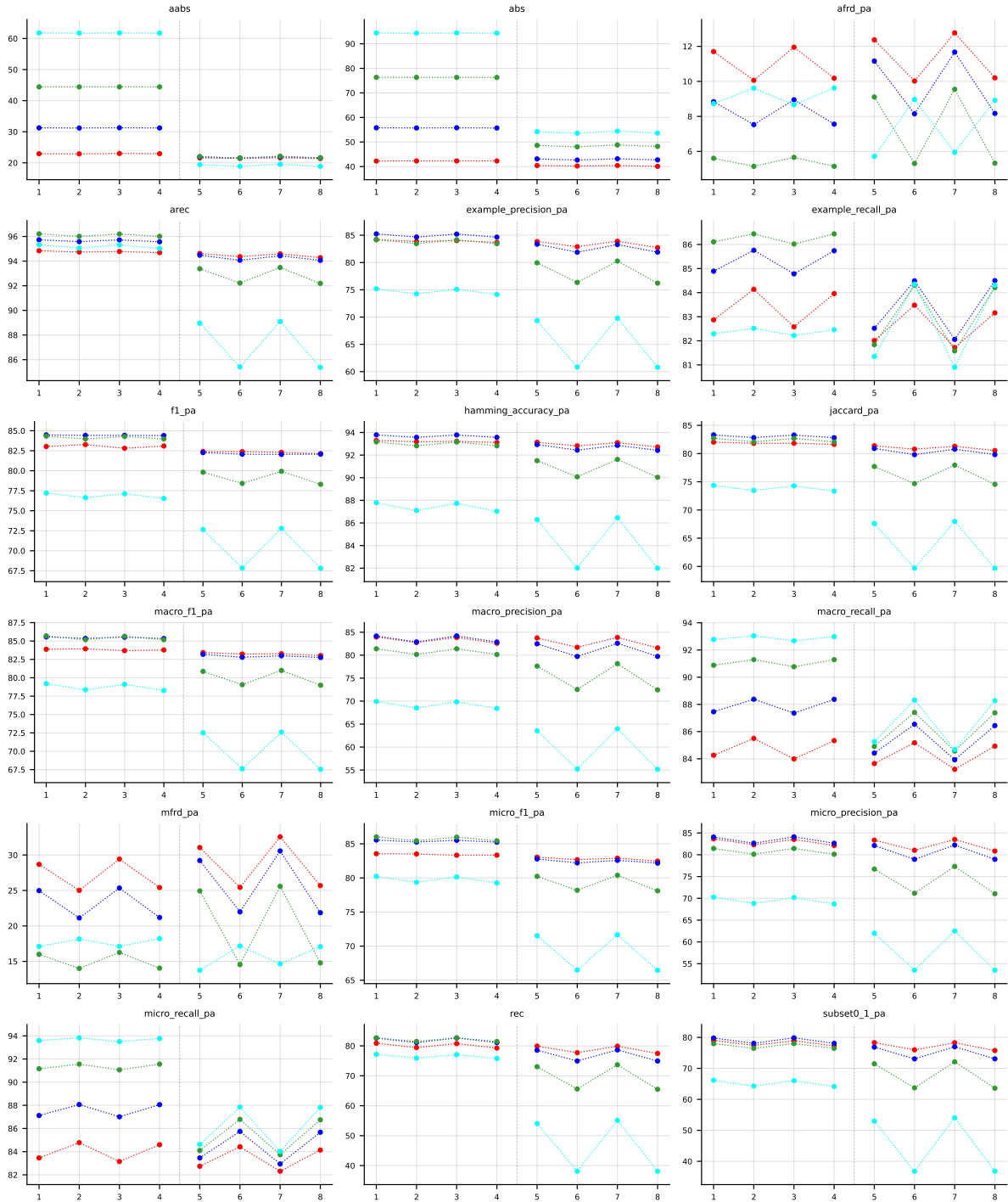


Table 12: Partial abstention (PartialAbstention) on scene (RF base learner).

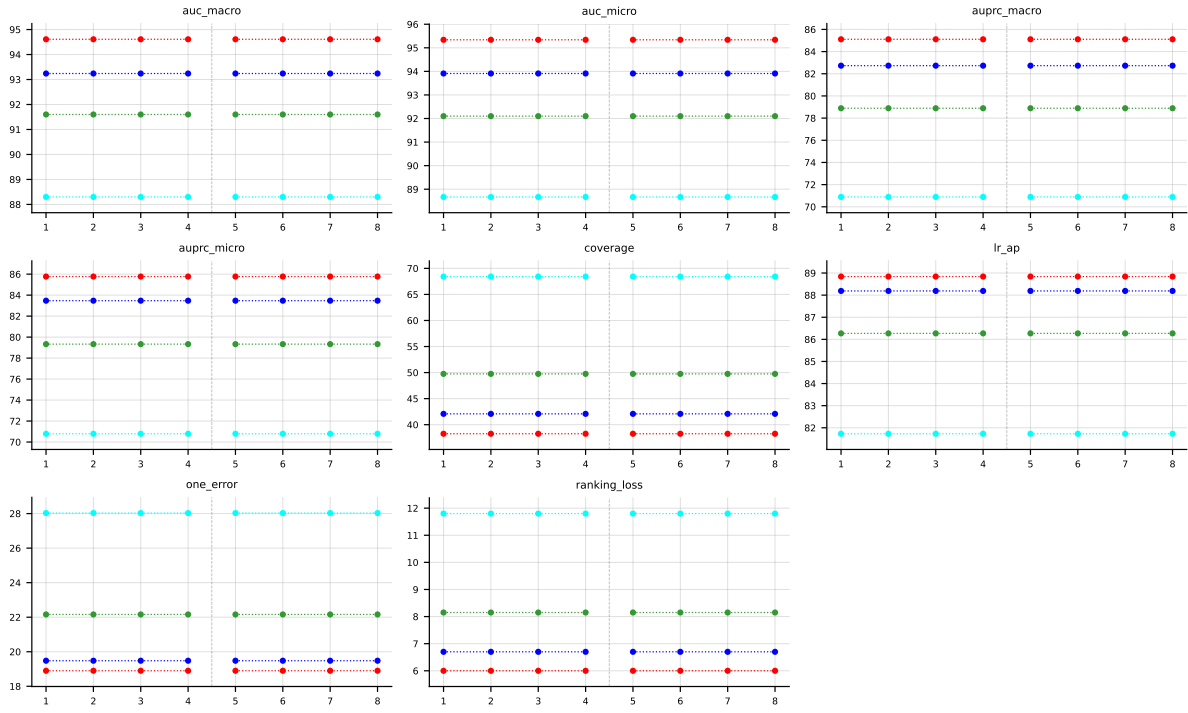


Table 13: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on **scene** (RF base learner).

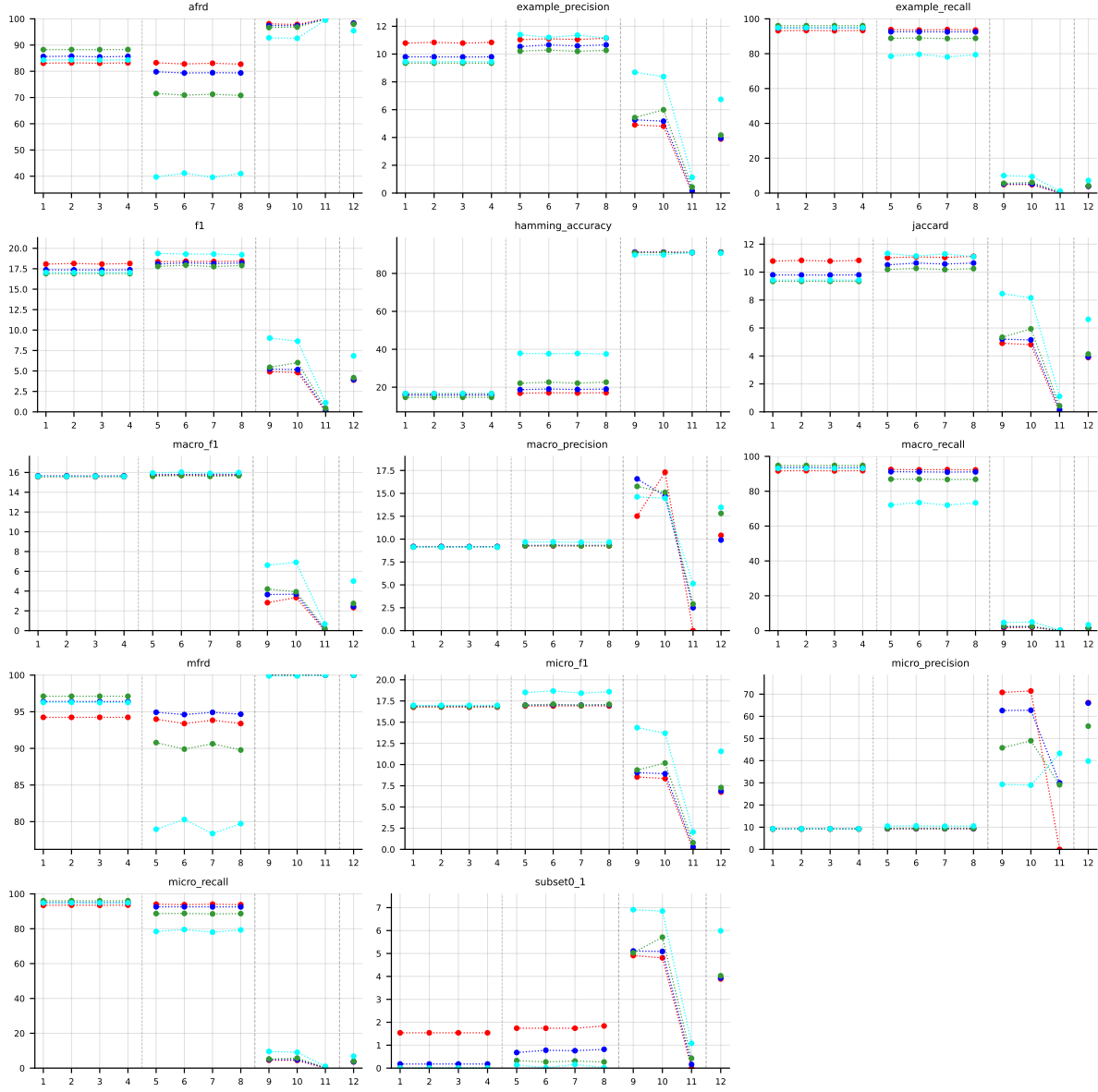


Table 14: Standard MLC predictions (BinaryVector) on PlantPseAAC (RF base learner).

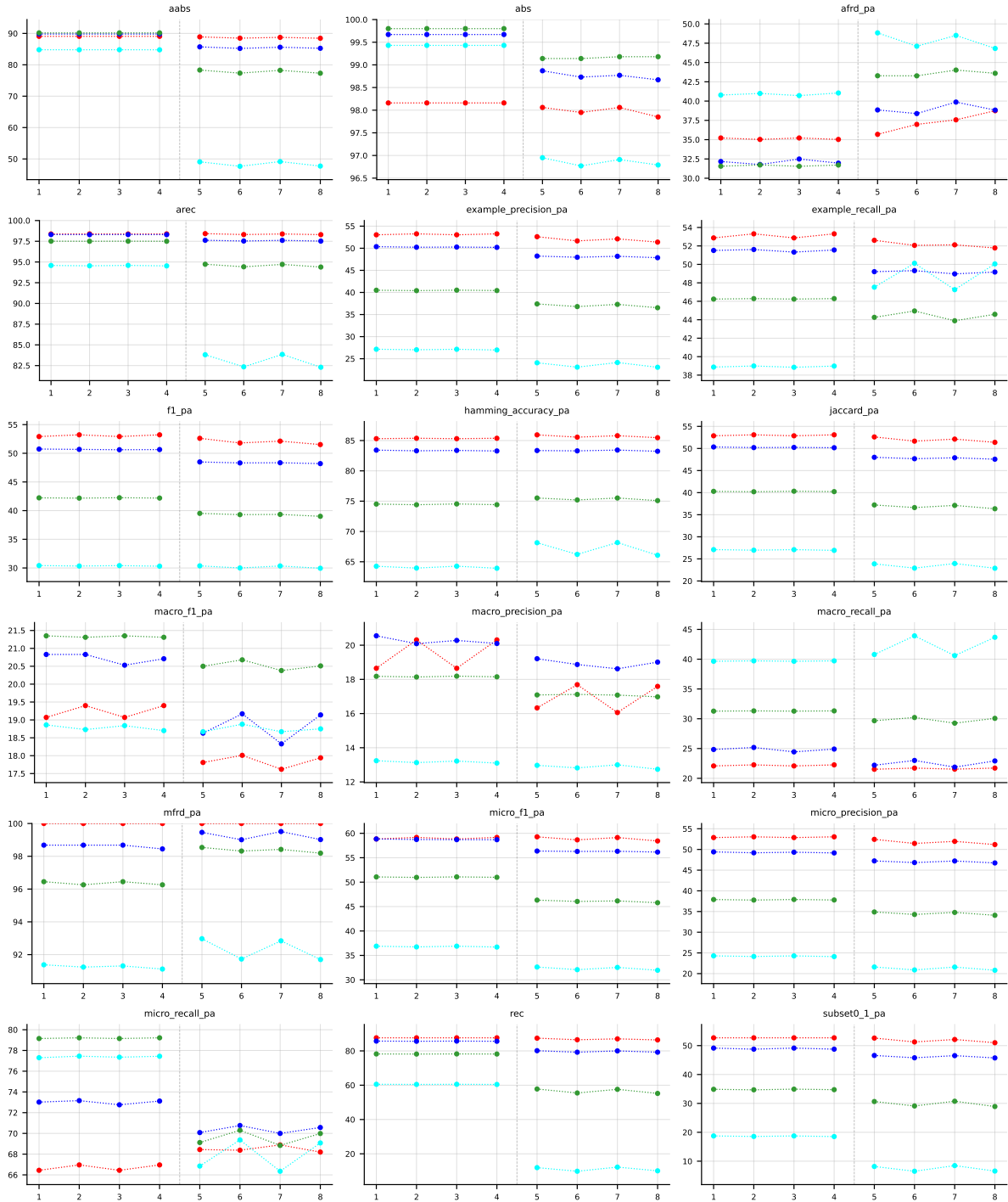


Table 15: Partial abstention (PartialAbstention) on PlantPseAAC (RF base learner).

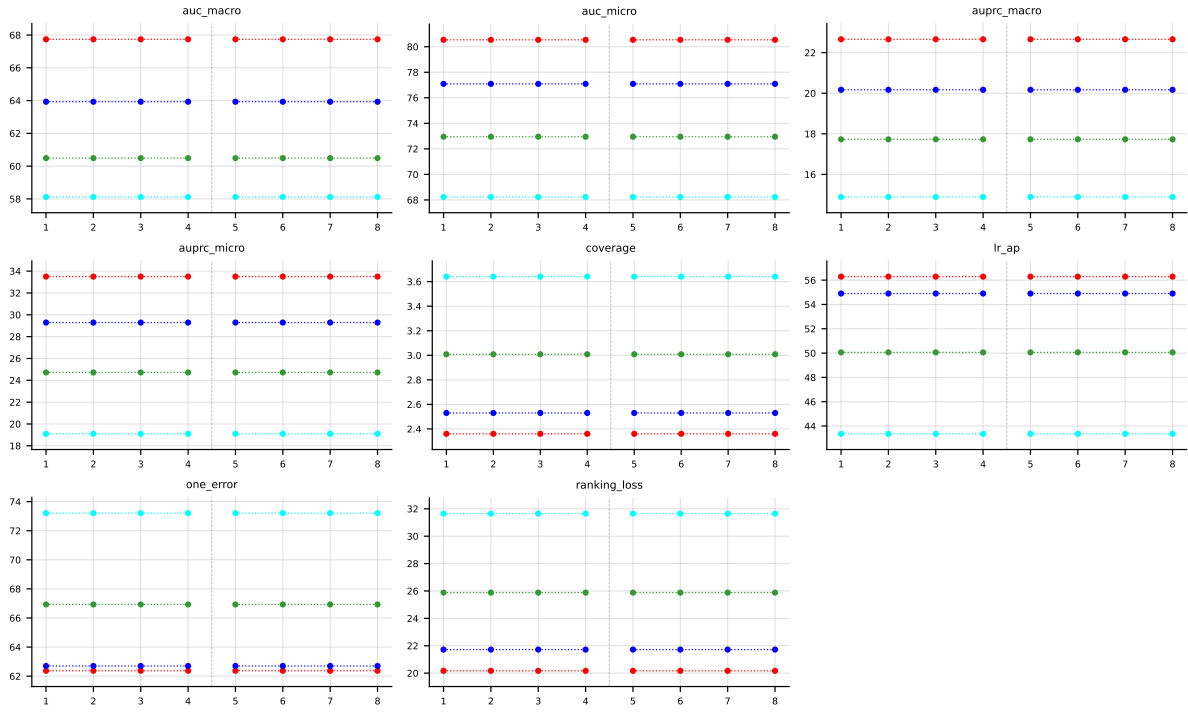


Table 16: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on PlantPseAAC (RF base learner).

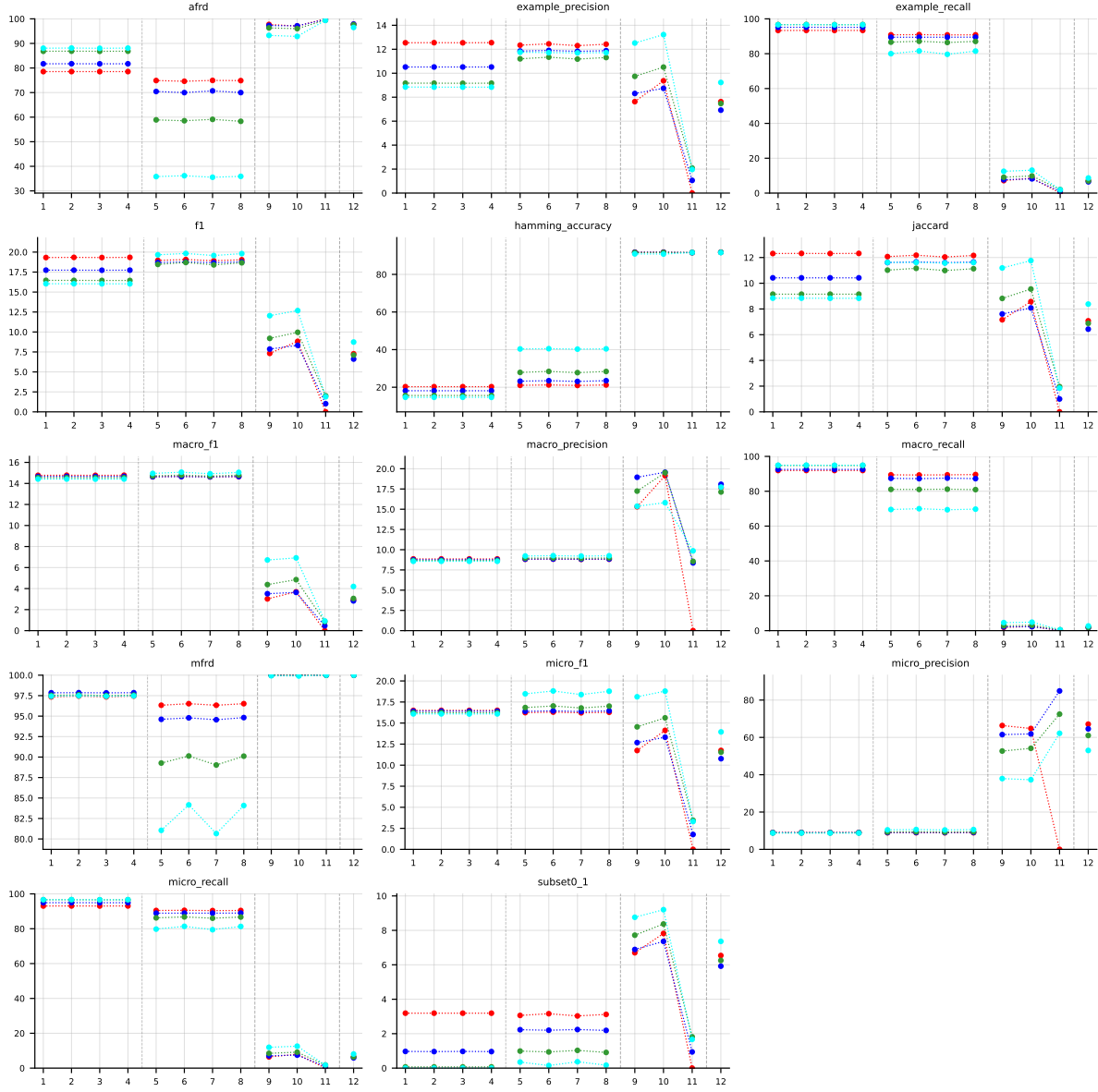


Table 17: Standard MLC predictions (BinaryVector) on HumanPseAAC (RF base learner).

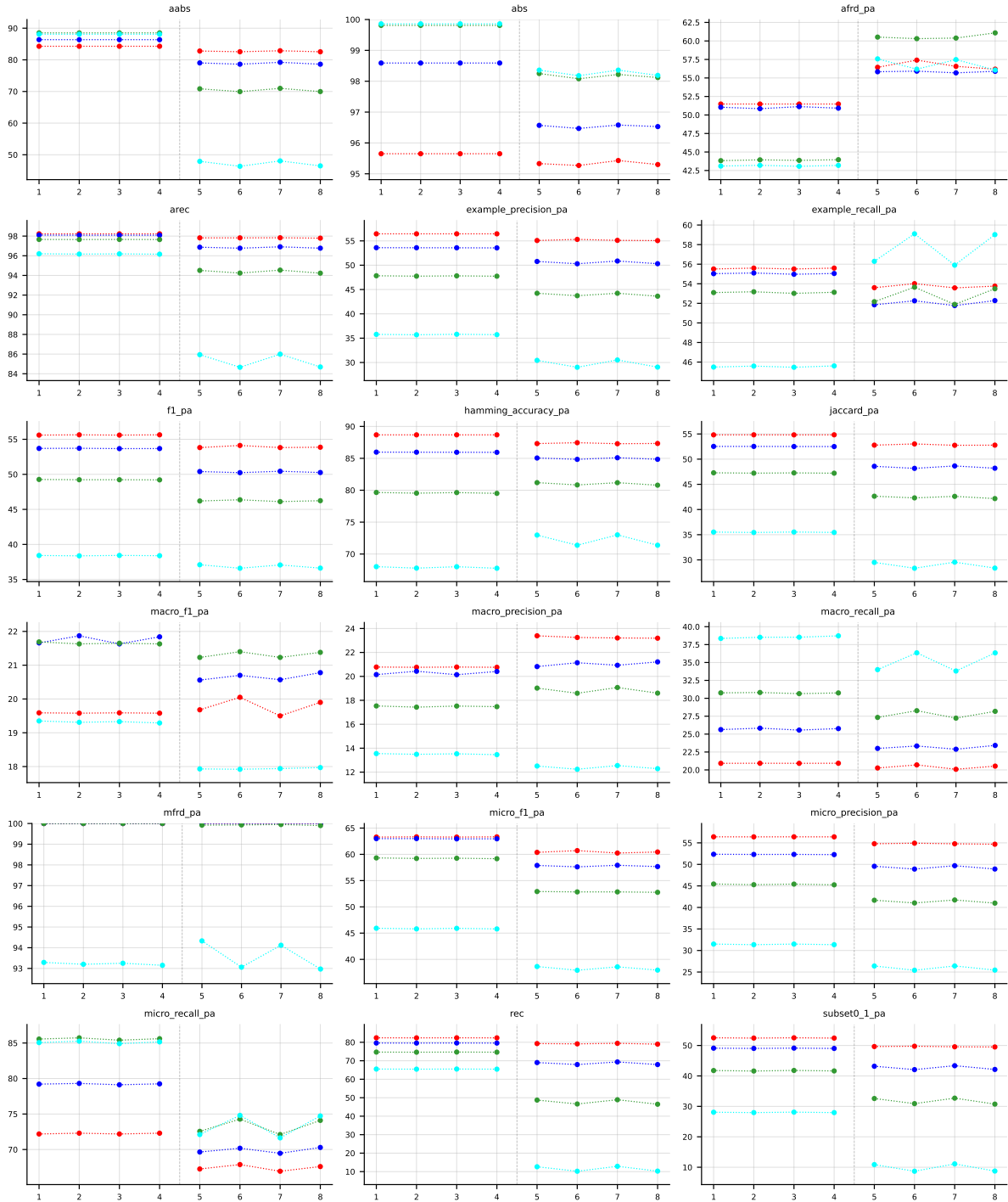


Table 18: Partial abstention (PartialAbstention) on HumanPseAAC (RF base learner).

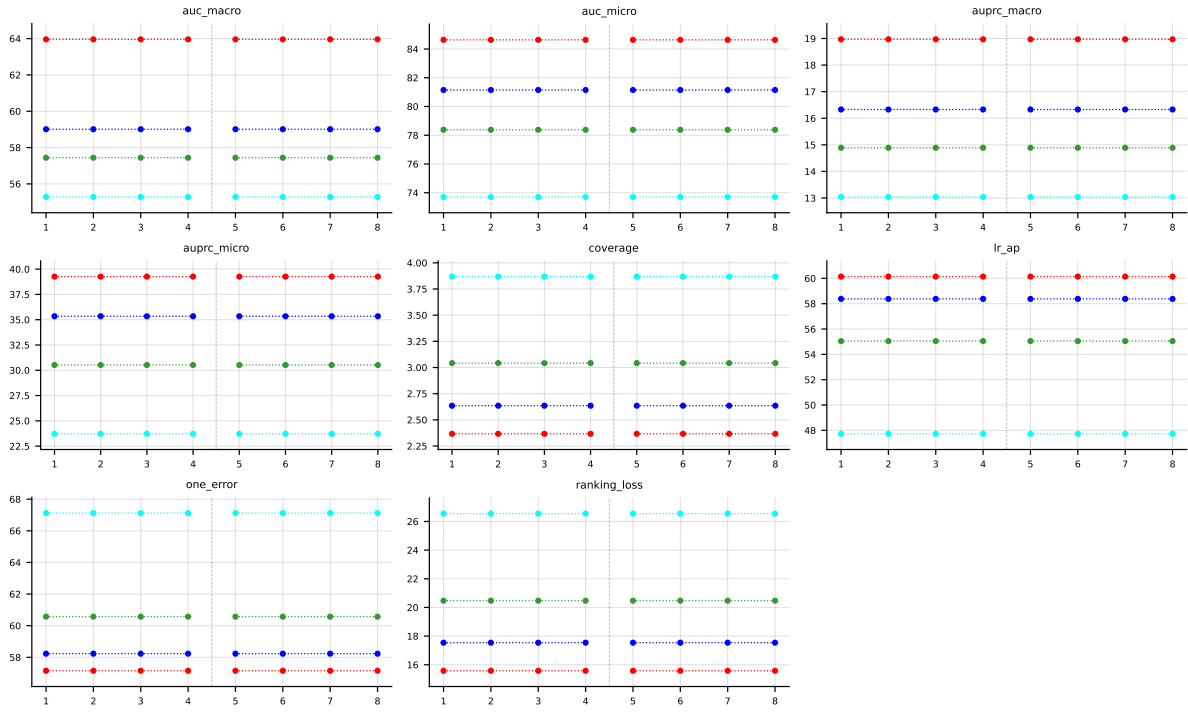


Table 19: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on HumanPseAAC (RF base learner).

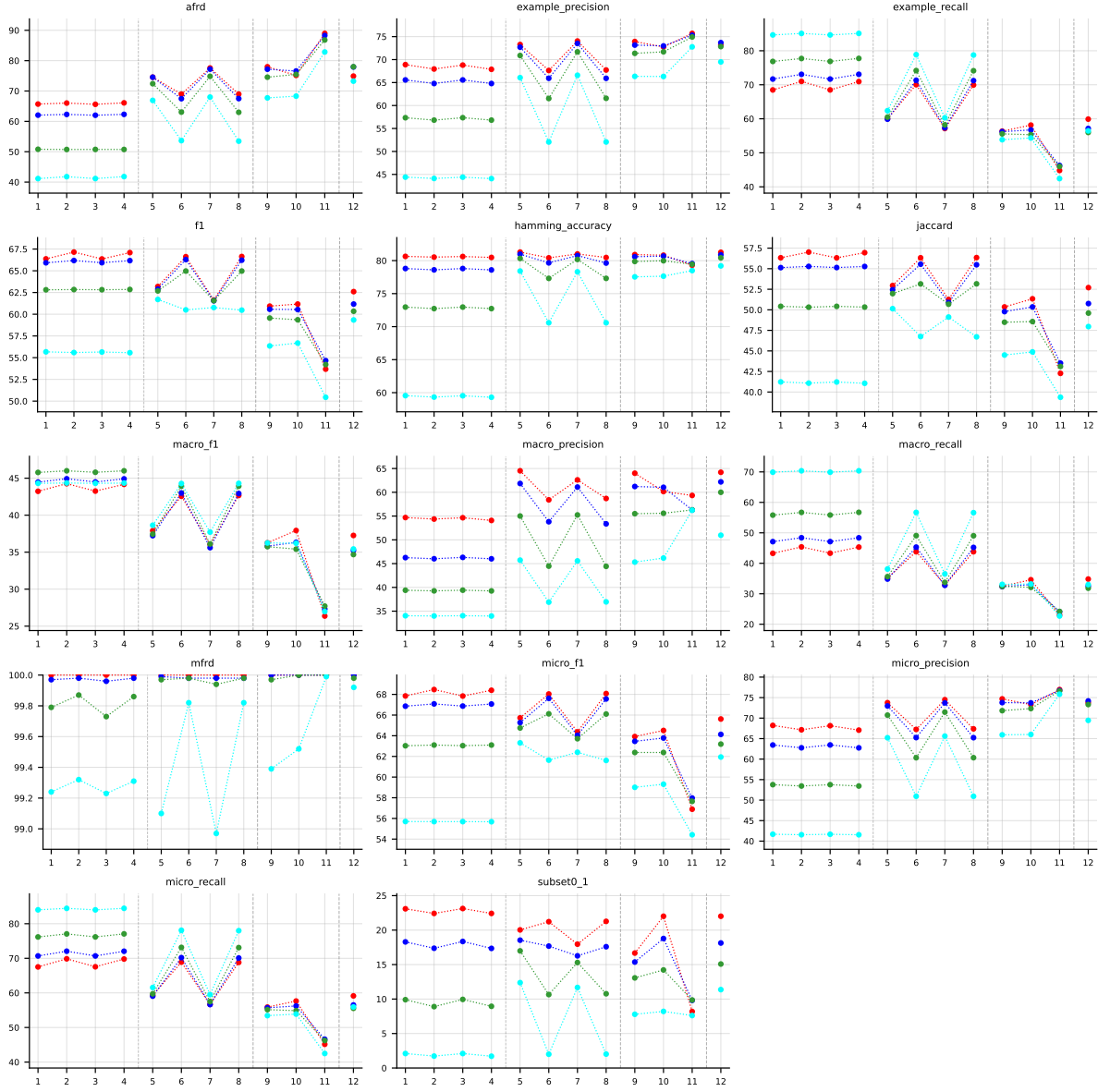


Table 20: Standard MLC predictions (BinaryVector) on **Yeast** (RF base learner).

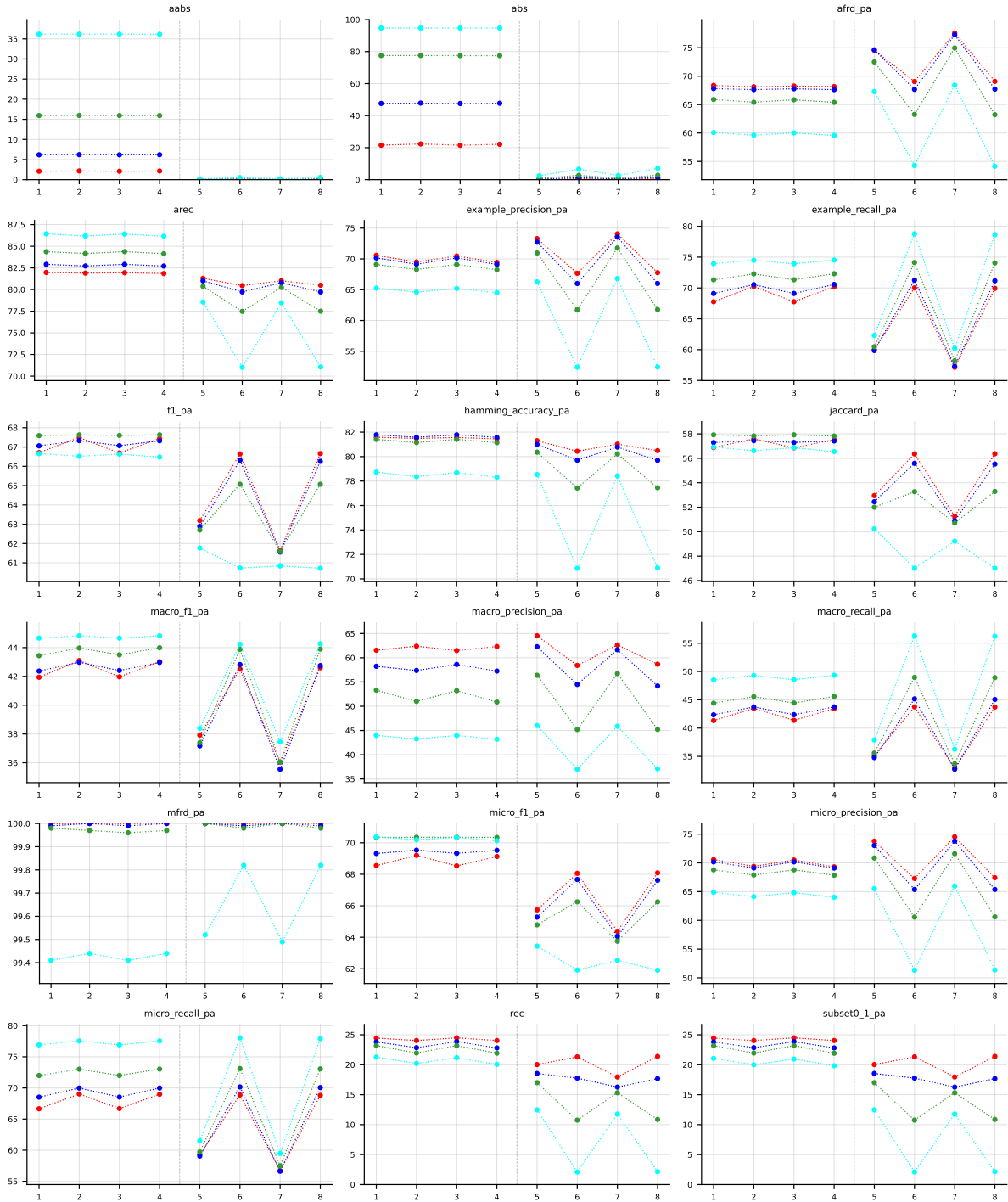


Table 21: Partial abstention (PartialAbstention) on Yeast (RF base learner).

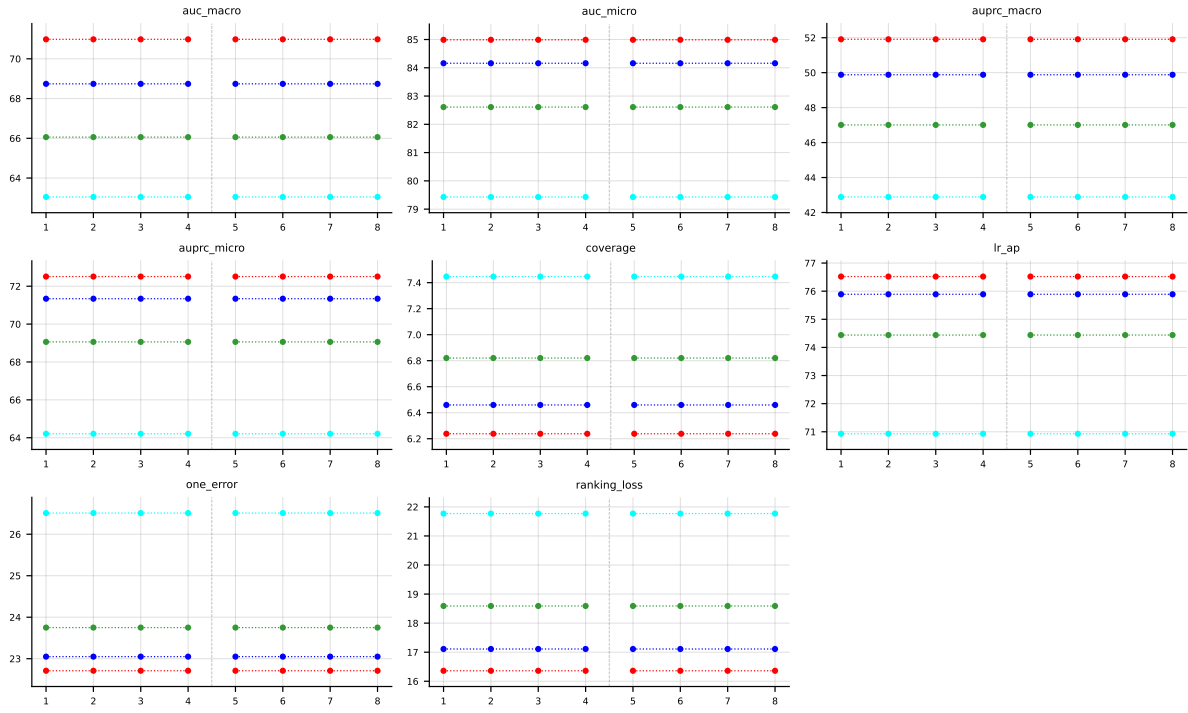


Table 22: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on Yeast (RF base learner).

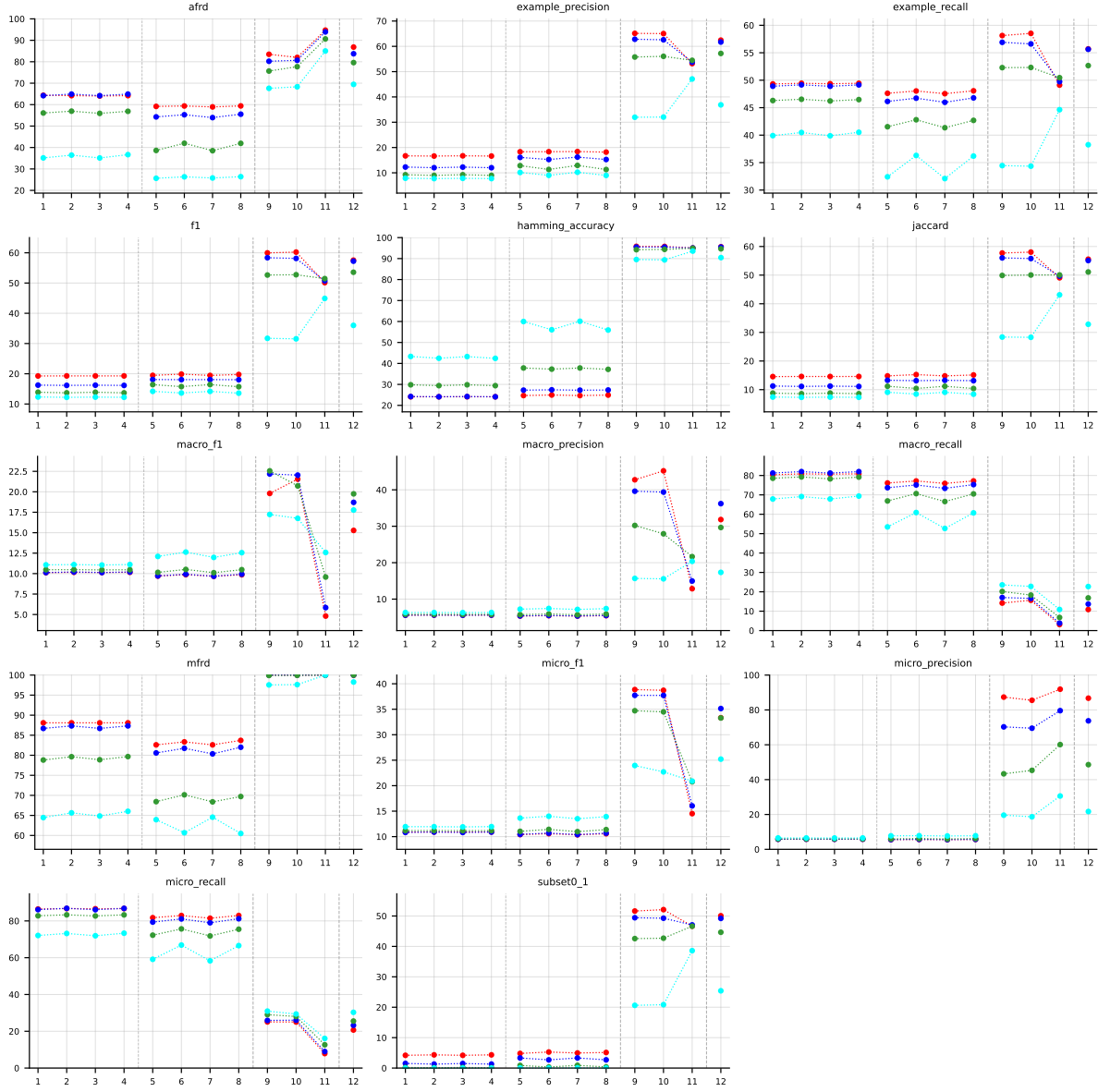


Table 23: Standard MLC predictions (BinaryVector) on birds (RF base learner).

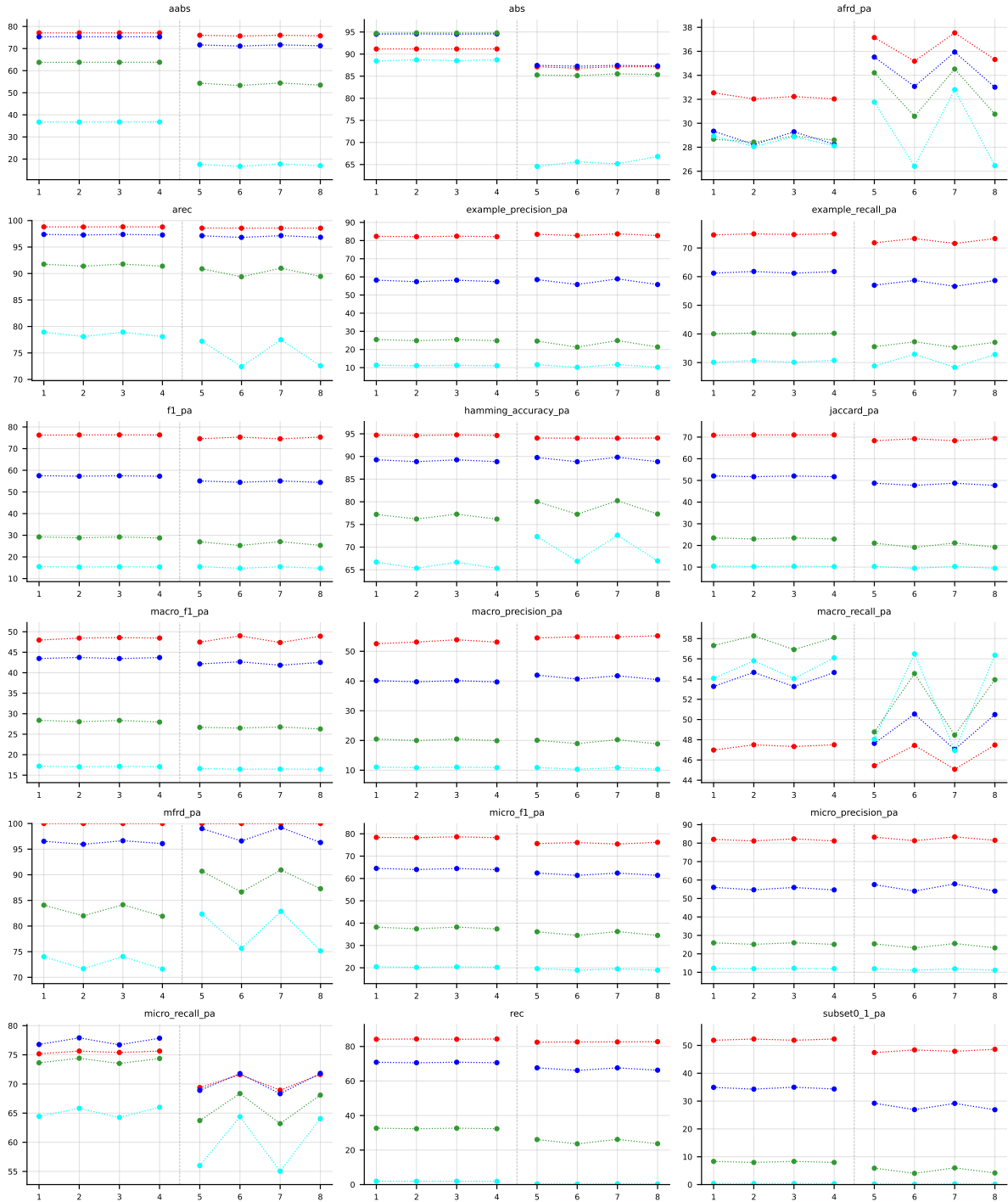


Table 24: Partial abstention (PartialAbstention) on **birds** (RF base learner).

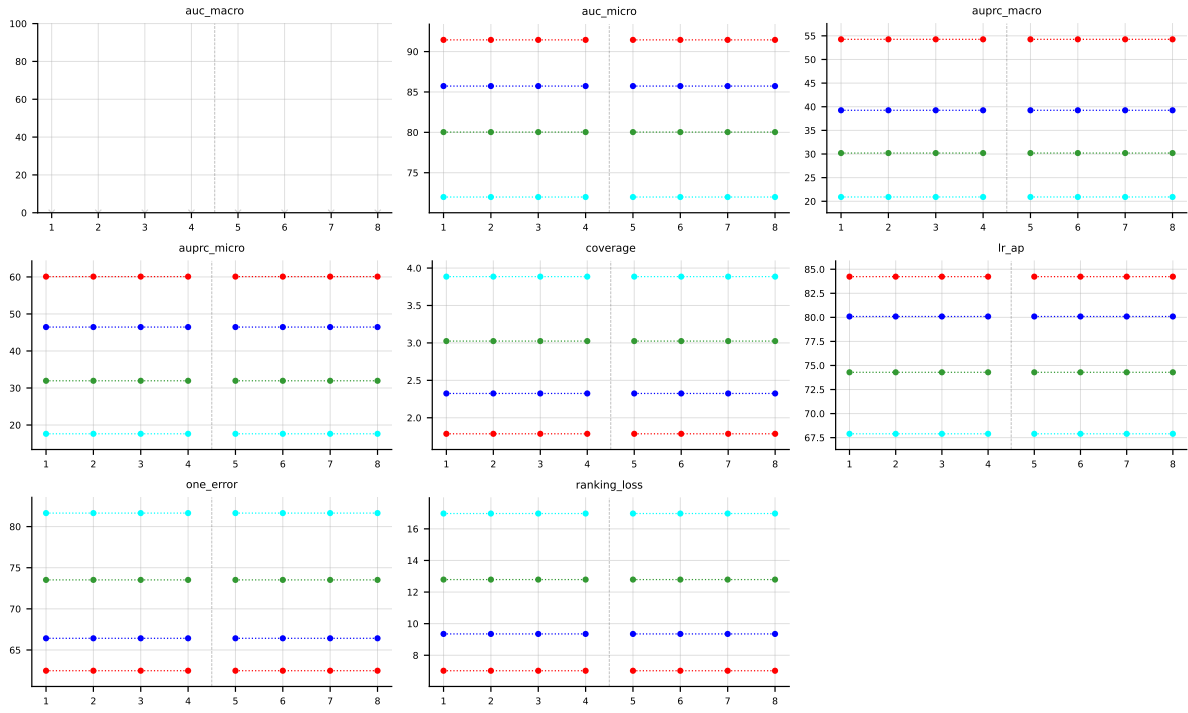


Table 25: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `birds` (RF base learner).

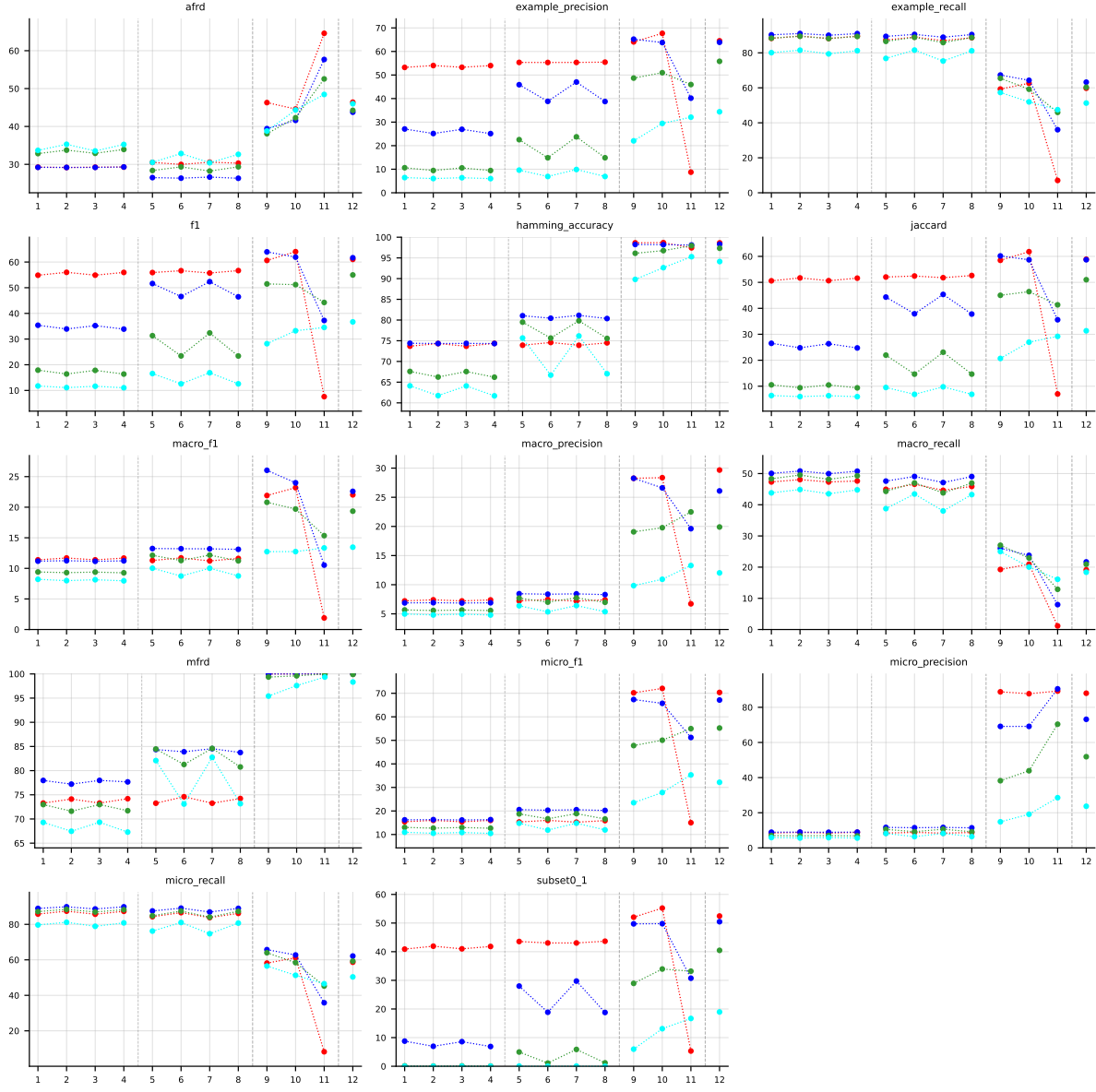


Table 26: Standard MLC predictions (BinaryVector) on `medical` (RF base learner).

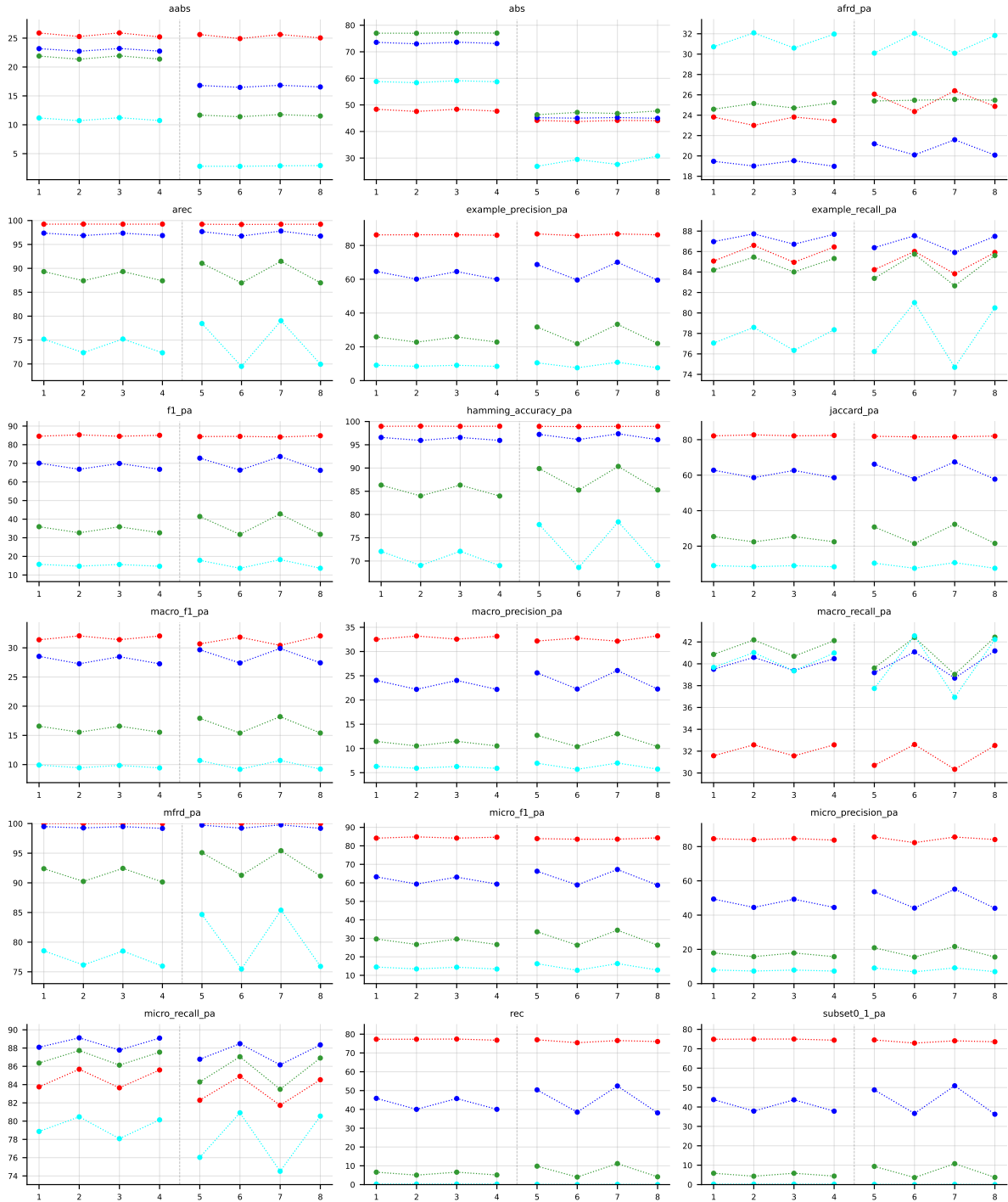


Table 27: Partial abstention (PartialAbstention) on medical (RF base learner).

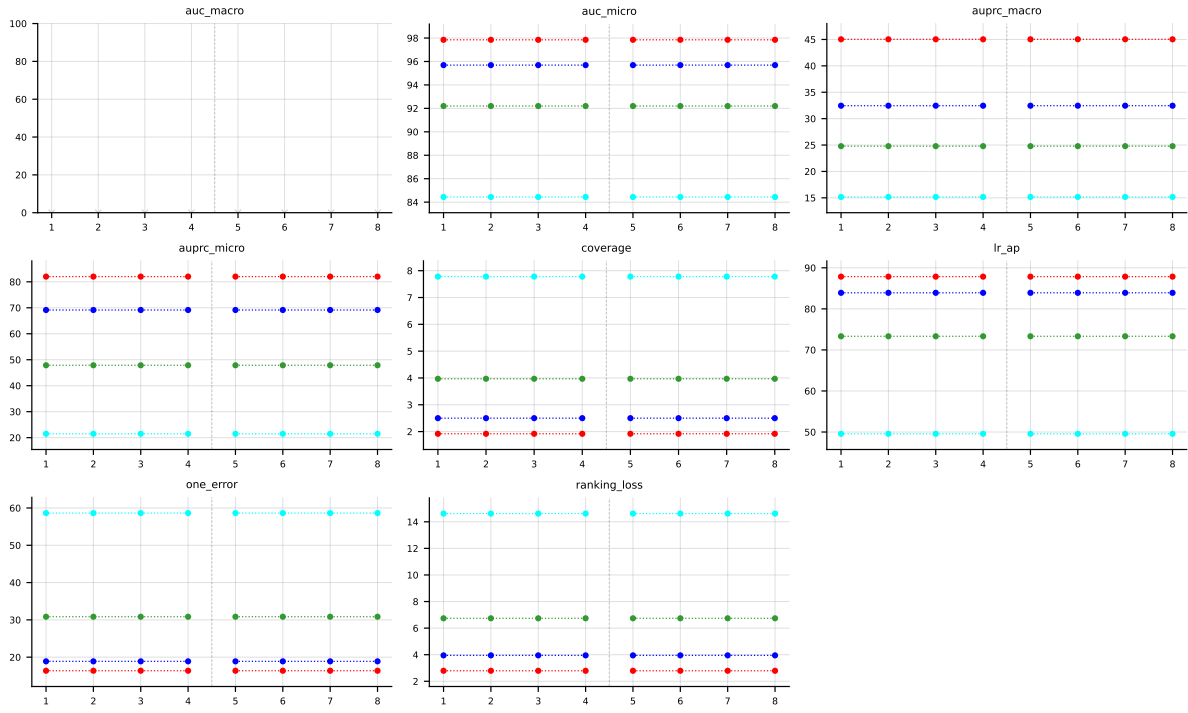


Table 28: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on medical (RF base learner).

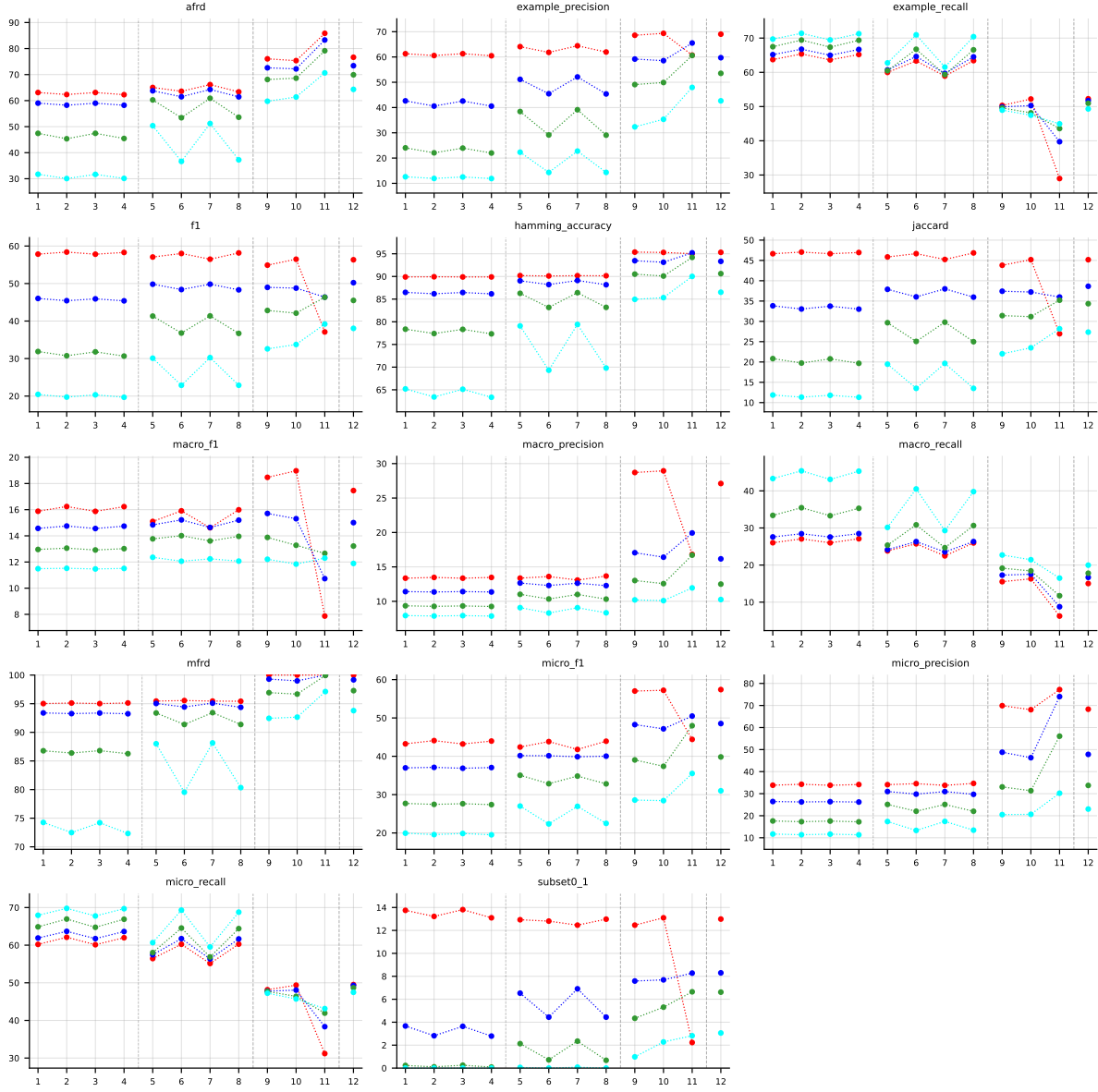


Table 29: Standard MLC predictions (BinaryVector) on **enron** (RF base learner).

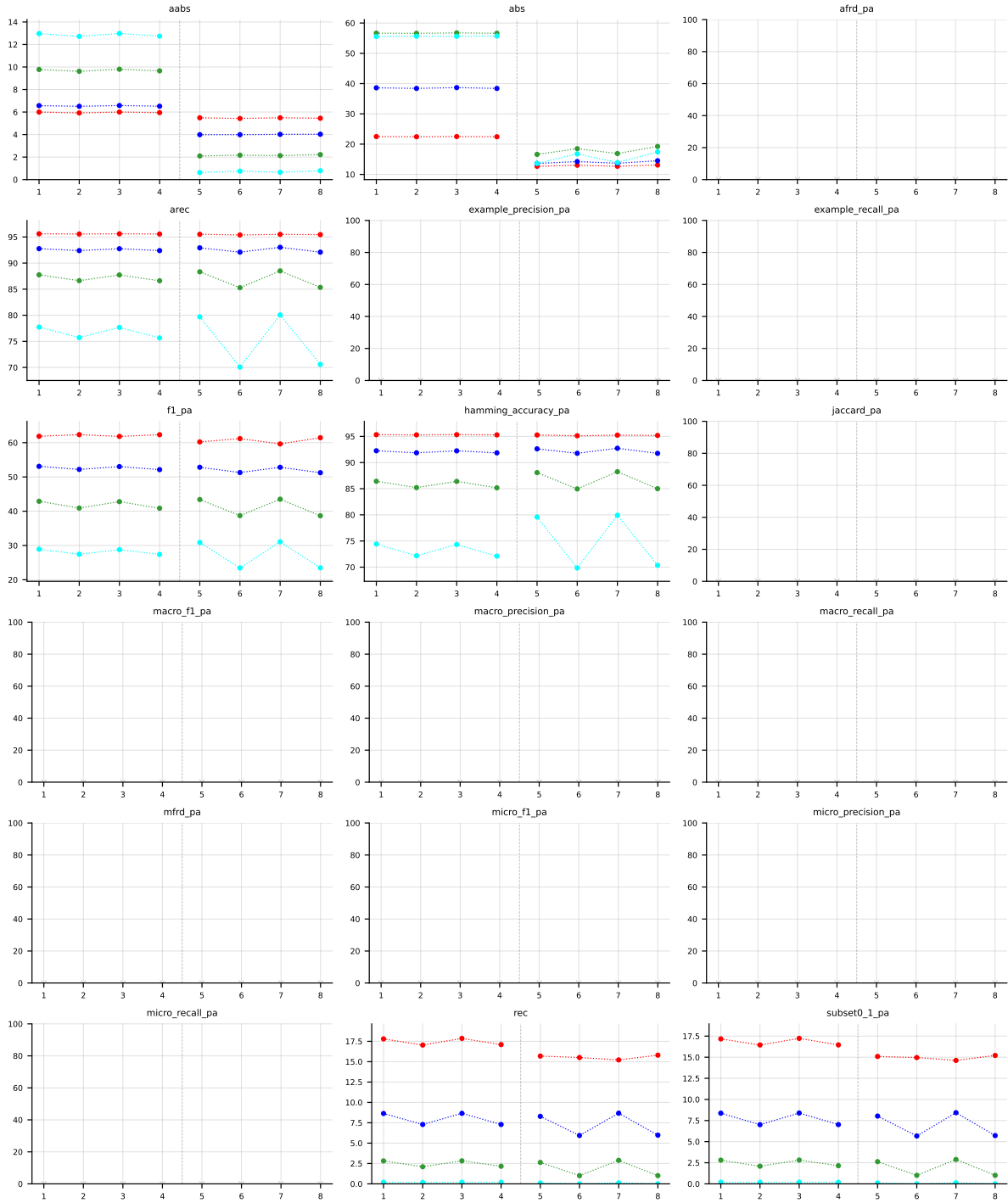


Table 30: Partial abstention (PartialAbstention) on **enron** (RF base learner).

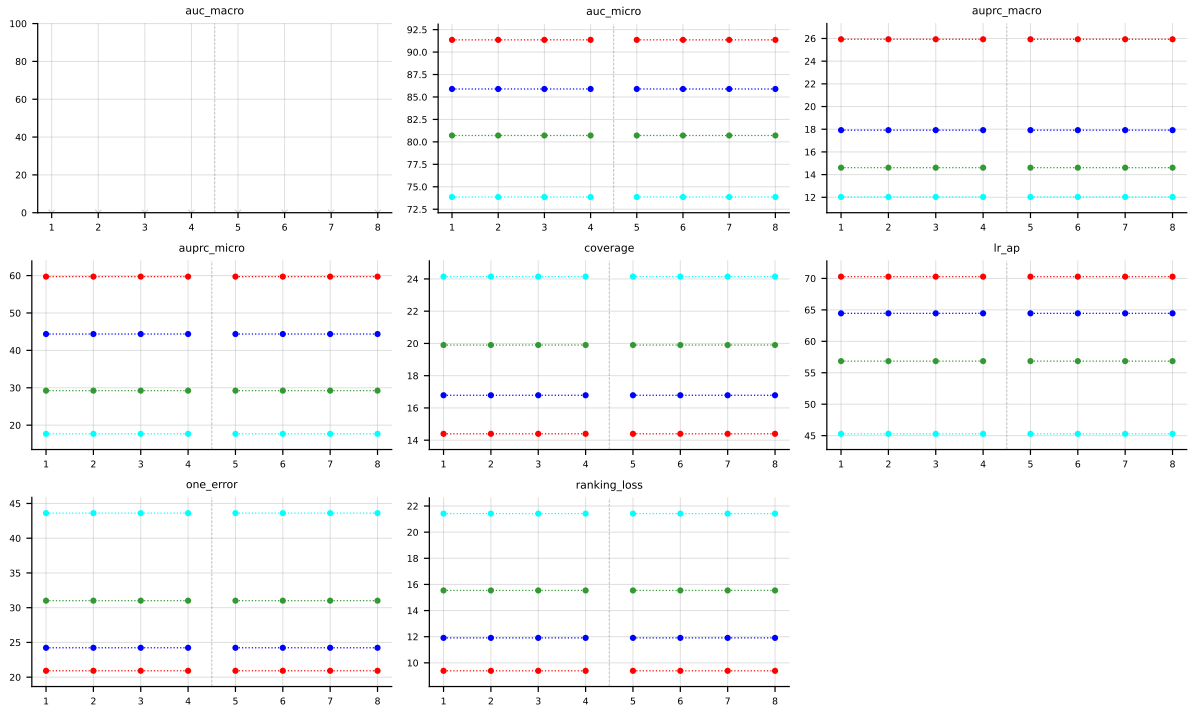


Table 31: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on **enron** (RF base learner).

B LGBM aggregated results

LGBM • Aggregate • BinaryVector — Standard MLC predictions

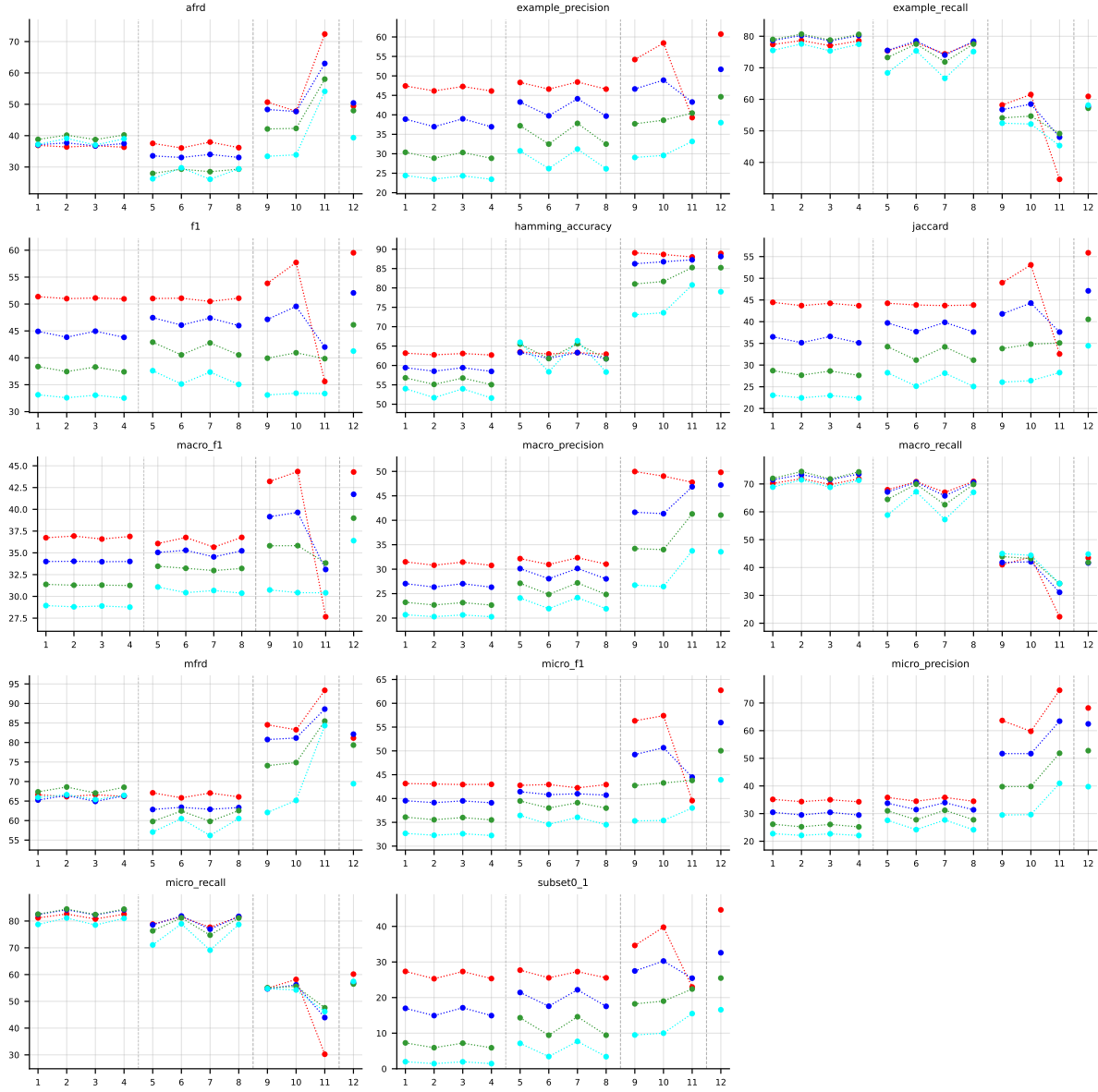


Table 32: Standard MLC predictions (BinaryVector), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.

LGBM • Aggregate • PartialAbstention — Partial abstention

LGBM • Aggregate • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

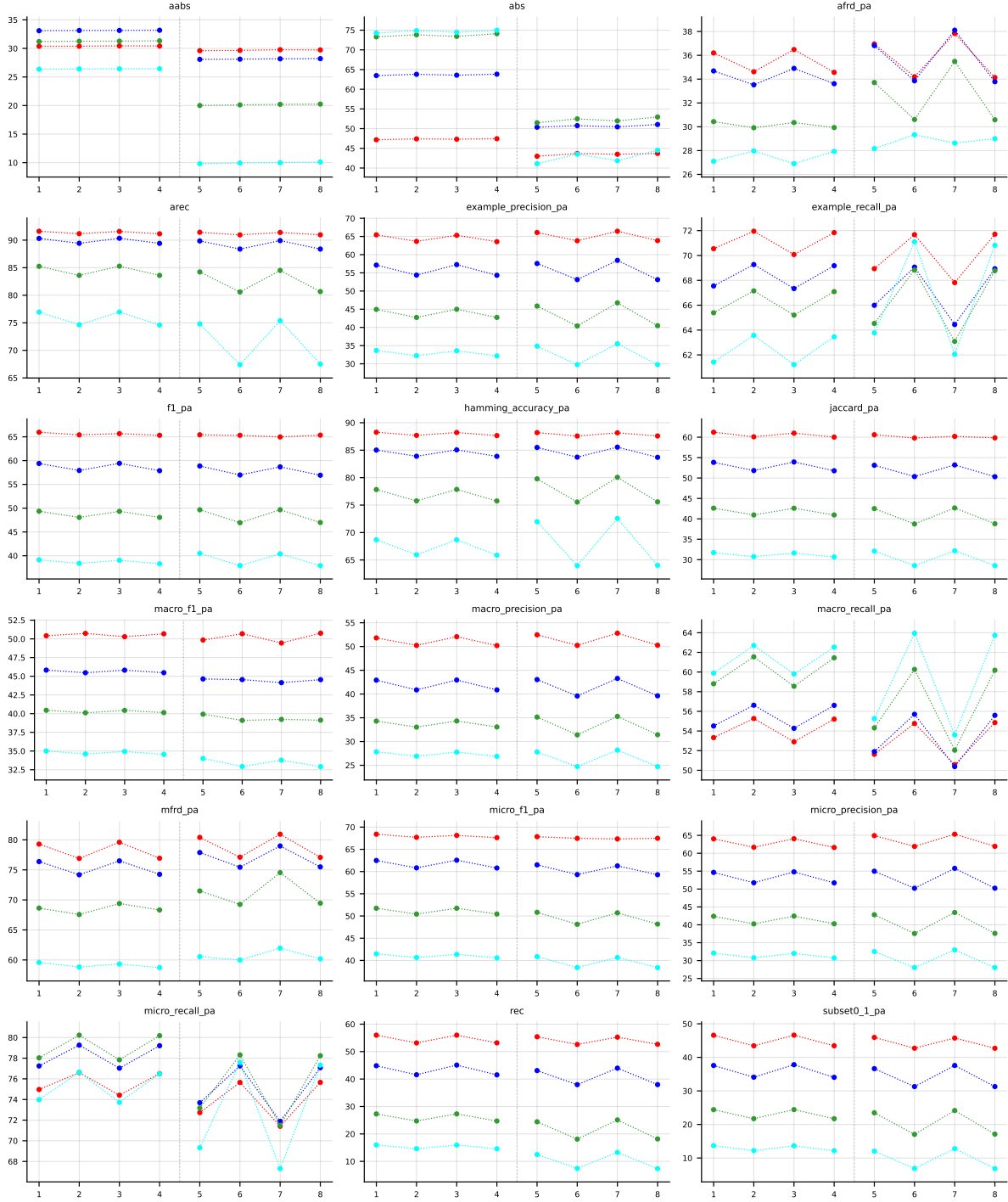


Table 33: Partial abstention (PartialAbstention), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.

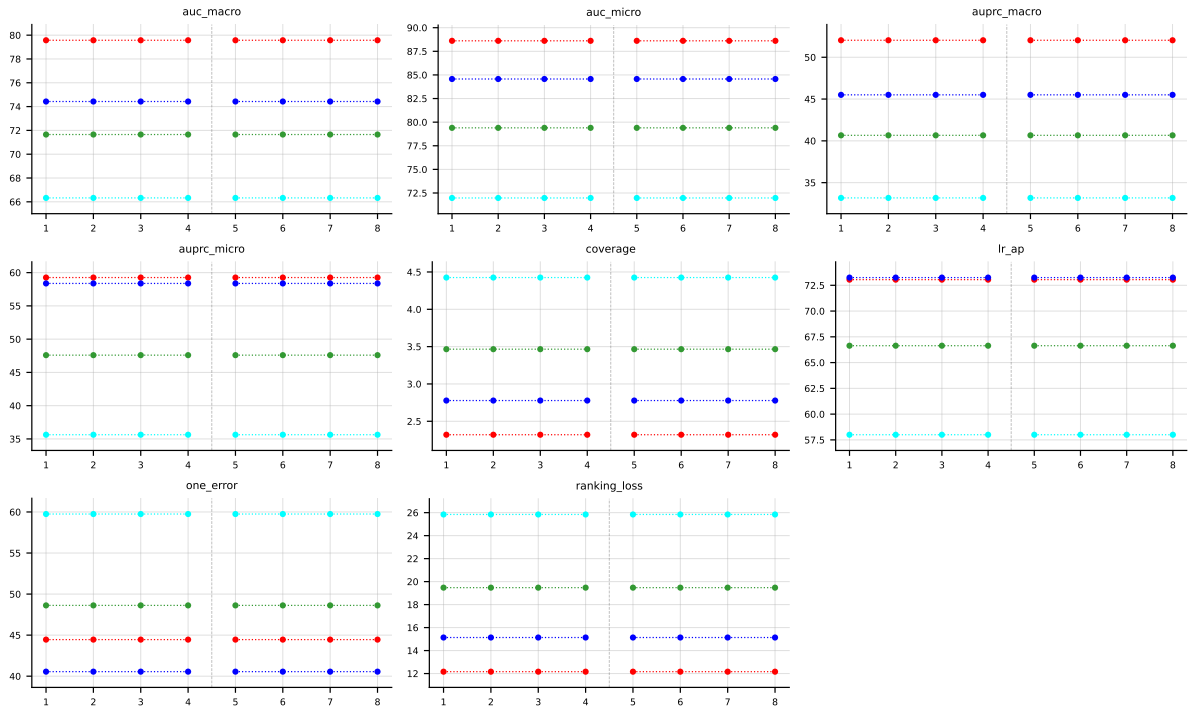


Table 34: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.